

CE/CZ4171

**Internet of Things:
Communications and Networking**

**IoT Wireless
Communications**



Outline

- Introduction to Wireless Communications
- Medium Access Control

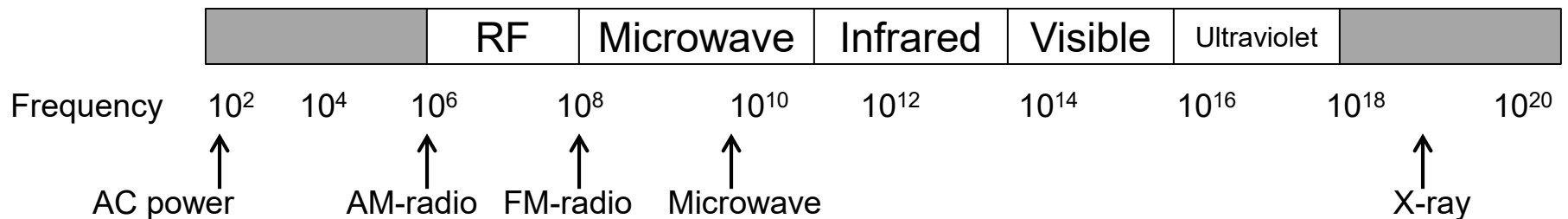


INTRODUCTION TO WIRELESS COMMUNICATIONS



What is Wireless Communications?

- Any network of communicating nodes that does not require the transmitter and receiver to be in physical contact
- Uses electromagnetic wave propagated through free-space, e.g.,
 - RF, Microwave, InfraRed, Lightwave



Possible Communication Types (Depending on the directions of communications allowed):

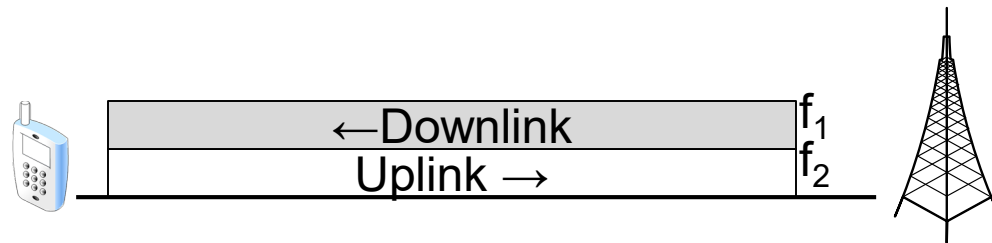
- **Simplex**: one-way communication (e.g., radio, TV, remote controller)
- **Half-duplex**: two-way communication but not simultaneous (e.g., push-to-talk radios)
- **Full-duplex**: two-way communication (e.g., cellular phones)
 - Frequency-division duplex (FDD)
 - Time-division duplex (TDD): simulated full-duplex

Note: Applicable to both wired and wireless

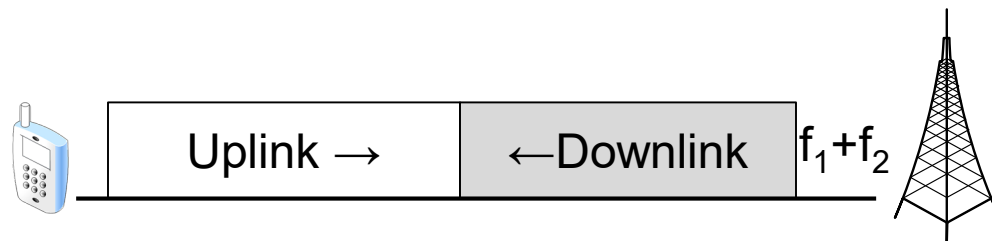


Possible Communication Types (Depending on the directions of communications allowed):

- **Full-duplex**
 - Frequency-division duplex (FDD)



- Time-division duplex (TDD): simulated full-duplex



Advantages of Wireless Communication

- Provides mobility
 - Mobility = ability to move freely, easily move
 - A user can send or receive a message no matter where he or she is located
- Added convenience/reduced cost
 - Enables communication without installing an expensive infrastructure
 - Can easily set-up temporary LANs
 - Disaster situations
 - Office moves
 - Developing nations utilize cellular telephony rather than laying twisted-pair wires to each home
- Use resources only when sending or receiving a signal



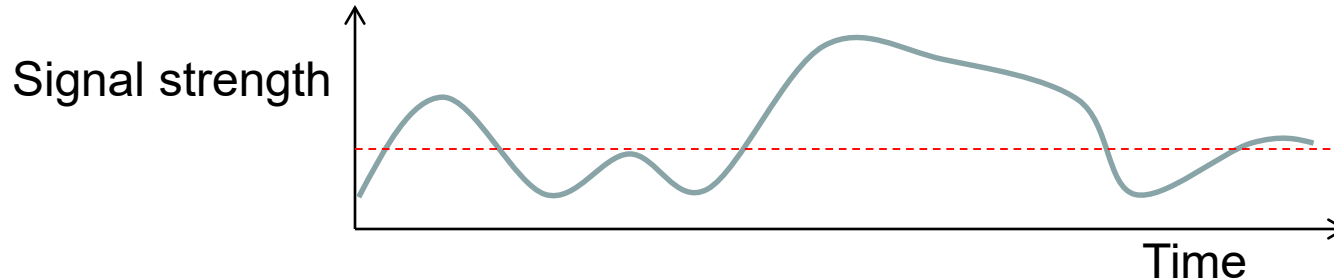
Summary of Special Requirements of Wireless

- Generally, mobility stresses all resources further:
 - CPU
 - Power
 - Bandwidth
 - Delay tolerance (realtime multimedia)
 - Radio spectrum
 - Human attention (more personalized, alert)
 - Physical size (small & compact)
 - Constraints on peripherals and GUIs (modality of interaction)

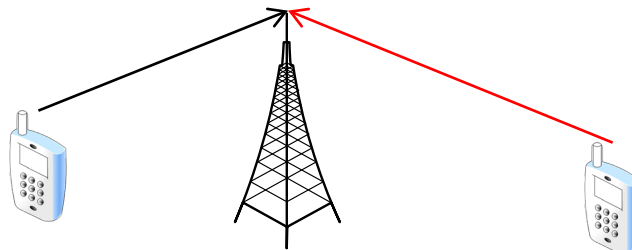


How Wireless differs from Wired

- Noisy, time-varying channel
 - BER (bit error rate) varies by orders of magnitude ($10^{-2} - 10^{-6}$)
 - Environmental conditions affect transmission (shadowing)



- Shared medium
 - Other users create interference
 - Must develop ways to share the channel



How Wireless differs from Wired

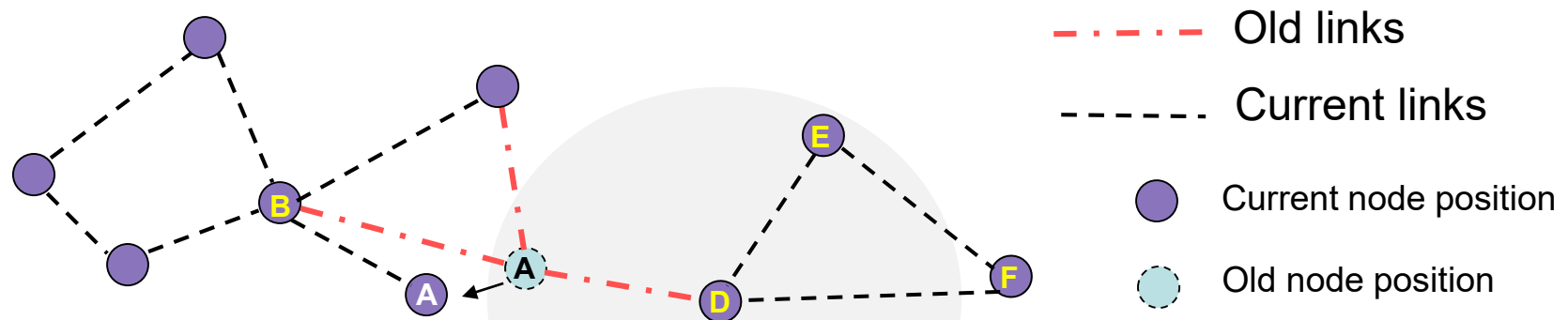
- Bandwidth is limited
 - FCC (Federal Communications Commission) determines how spectrum is allocated
 - ISM (industrial, scientific and medical) radio band for unlicensed use (902-928 MHz, 2.4-2.5 GHz and 5.725-5.875 GHz)
- Requires intelligent signal processing and communication to make efficient use of limited bandwidth in error-prone environment





How Wireless differs from Wired

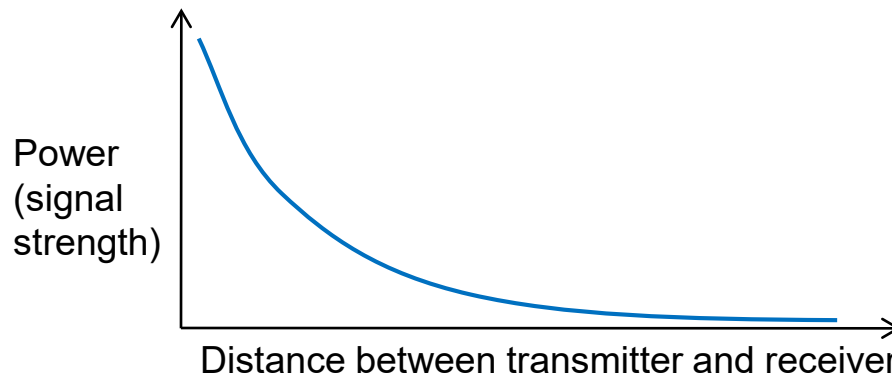
- Major differences due to broadcast nature of wireless communication
 - Transmitted signals can be received by an arbitrary (and perhaps unknown) number of other users
 - Cannot guarantee a link from every transmitting node to every intended receiving node



How Wireless differs from Wired

- Each transmitted message utilizes scarce resources (spectrum bandwidth)
 - Need to provide means for fair and efficient utilization of available bandwidth among transmitting nodes
- Transmitted signal power important parameter
 - Require enough signal power to reach destination node
 - Want to limit signal power to minimize interference and max battery life

contradictory

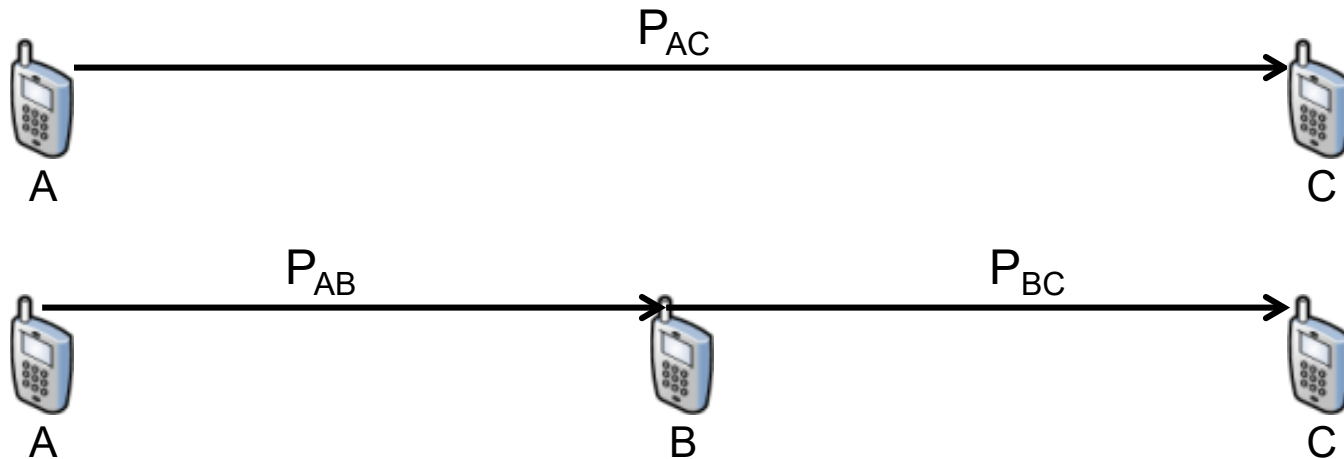


How Wireless differs from Wired

Receive_power = Transmit_power / (Distance^Path_loss_exponent)

Transmit_power = Receive_power * (Distance^Path_loss_exponent)

Path_loss_exponent > 2



Required Receive_power = 1 and Path_loss_exponent = 2

Distance from A to C is 100 (km), and B is located at the middle of A and C

For A to C, the required transmit power is: $1 * 100^2 = 10,000$

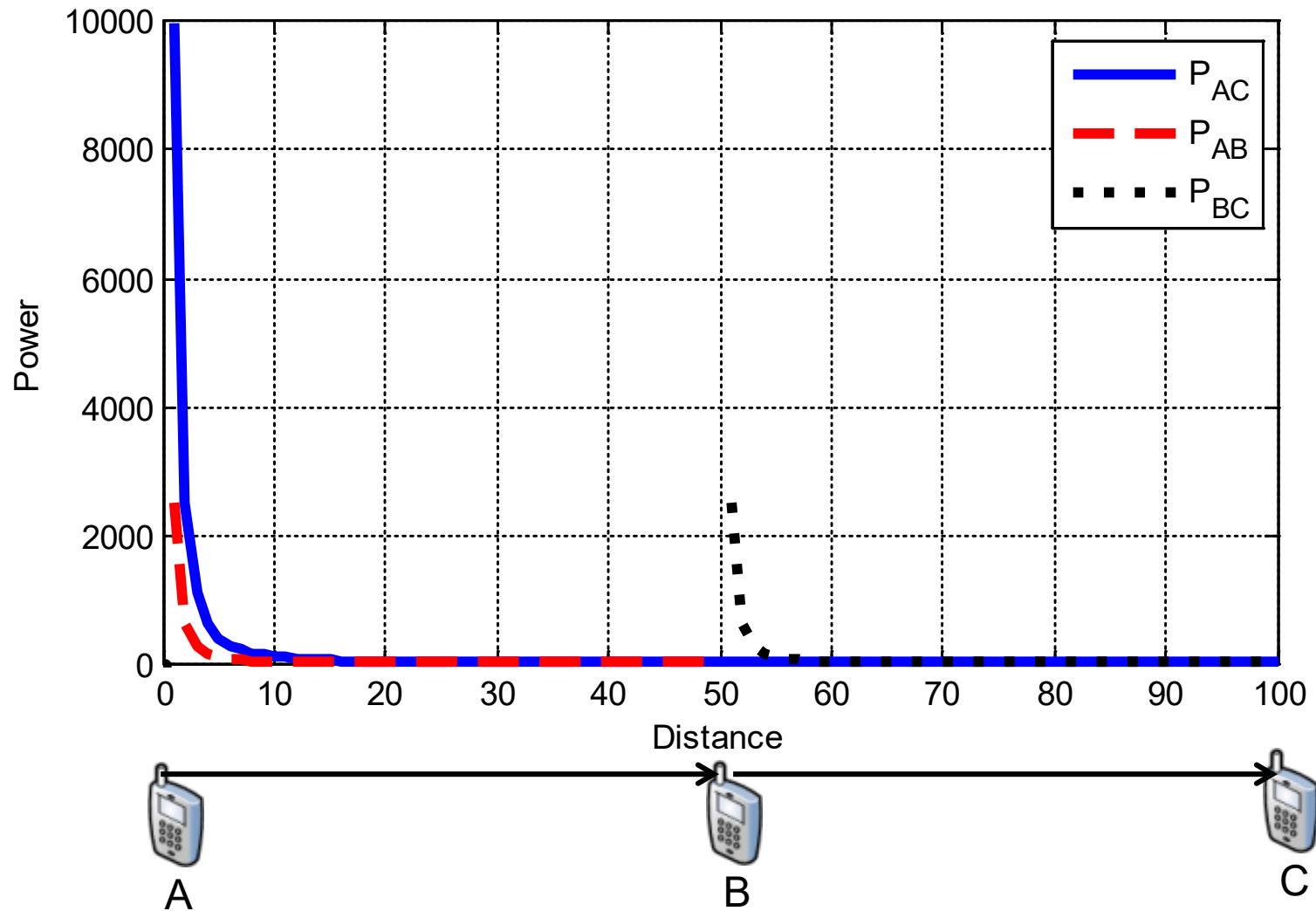
For A to B, the required transmit power is: $1 * 50^2 = 2,500$

For B to C, the required transmit power is: $1 * 50^2 = 2,500$

Therefore, $P_{AC} \gg P_{AB} + P_{BC}$



How Wireless differs from Wired



How Wireless differs from Wired

Environment	Path loss exponent
• Free space	2
• Urban area	2.7-3.5
• Suburban cellular radio	3-5
• Indoor line-of-sight	1.6-1.8
• Obstructed in building	4-6
• Obstructed in factory	2-3



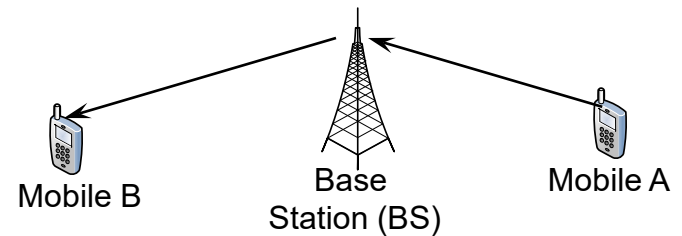
Network Topologies

- Types of network topologies

- Centralized
- De-centralized (peer-to-peer)
- Hybrid

- Centralized (hub-and-spoke) topology

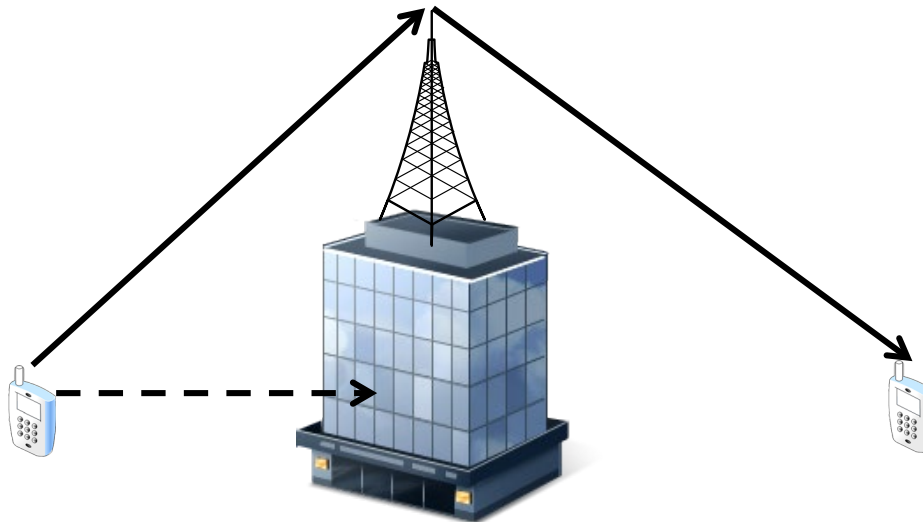
- Communication from one node to another goes through hub (base station or access point)
- Hub station controls nodes and monitors transmissions from each node
- Hub manages access by nodes to network's allocated bandwidth
- Configuration for cellular systems and many WLAN networks



Centralized Topology

Advantages

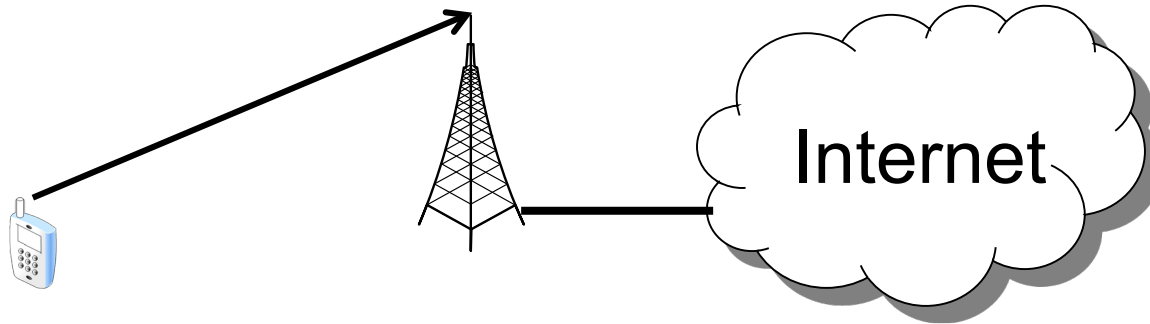
- Efficient use of transmit power
 - Compared with fully-connected peer-to-peer, nodes can reach other nodes twice the distance with same signal power (using high powered Base Station as intermediary)
- Hub/BS can be appropriately placed to minimize obstruction



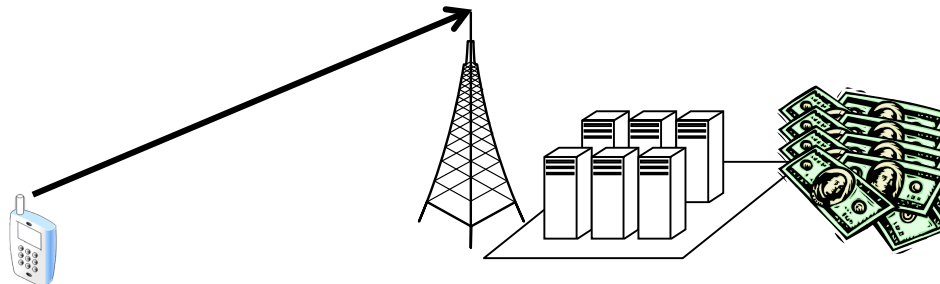
Centralized Topology

Advantages

- Hub/BS provides connection to backbone network → reason many WLANs have centralized topology



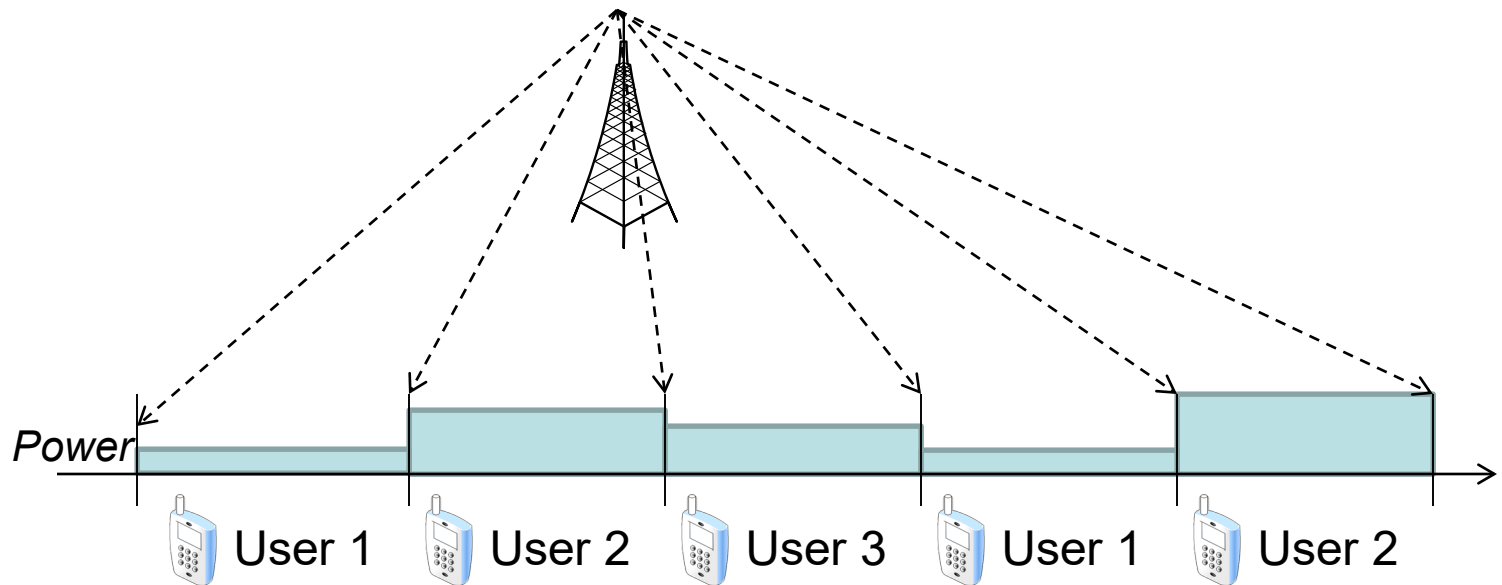
- Nodes can be made simple and BS complex



Centralized Topology

Advantages

- Helpful for power control → a central point can determine required power for nodes to minimize interference and conserve battery
- Hub can provide common timing reference



Centralized Topology

Disadvantages

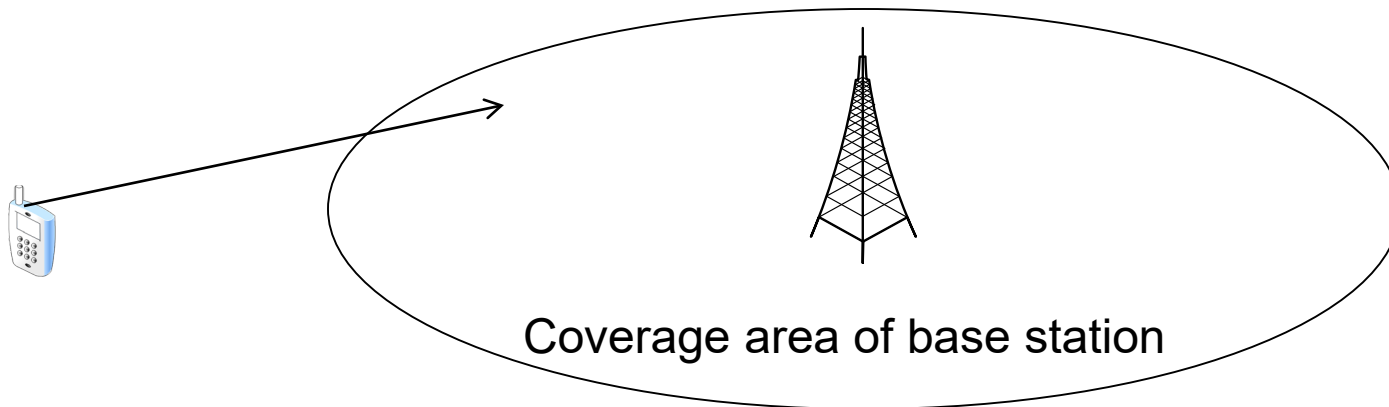
- Single point of failure
- Delay due to multiple transmit/receive operations
- Cannot deal with unpredictable propagation environments



Centralized Topology

Disadvantages

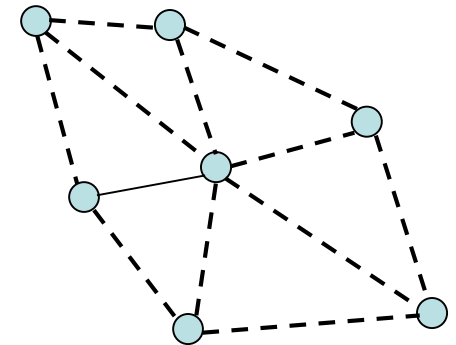
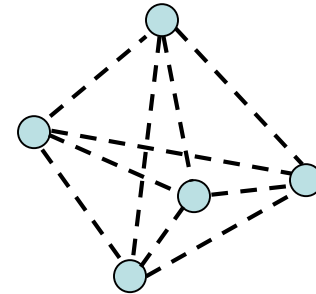
- Cannot cover wide areas where connections exceed range of a single link
- Not suitable for ad-hoc networks
- Requires significant infrastructure setup



Decentralized Topologies

Types of Decentralized (peer-to-peer) topology

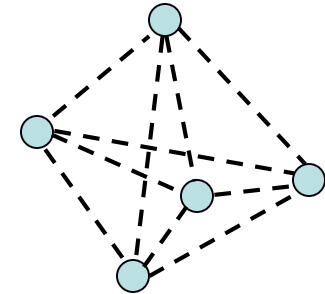
- **Fully-connected network**
(single hop network)
 - All nodes can communicate directly
 - Requires nodes to be co-located
- **Multi-hop network**
 - If nodes cannot directly reach destination, intermediate nodes must relay messages to destination
 - Widely used in ad-hoc networks where connectivity of all nodes is not guaranteed



Fully Connected peer-to-peer

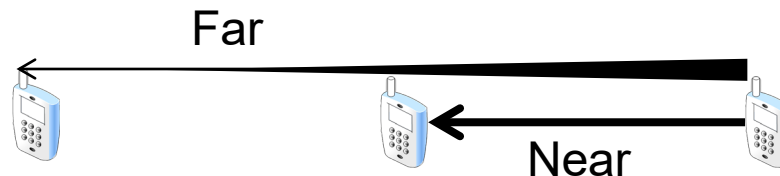
- Advantages

- No single point of failure
- No store-and-forward delay
- No routing → complexity of nodes reduced
- Can provide a node that is a gateway to backbone network



- Disadvantages

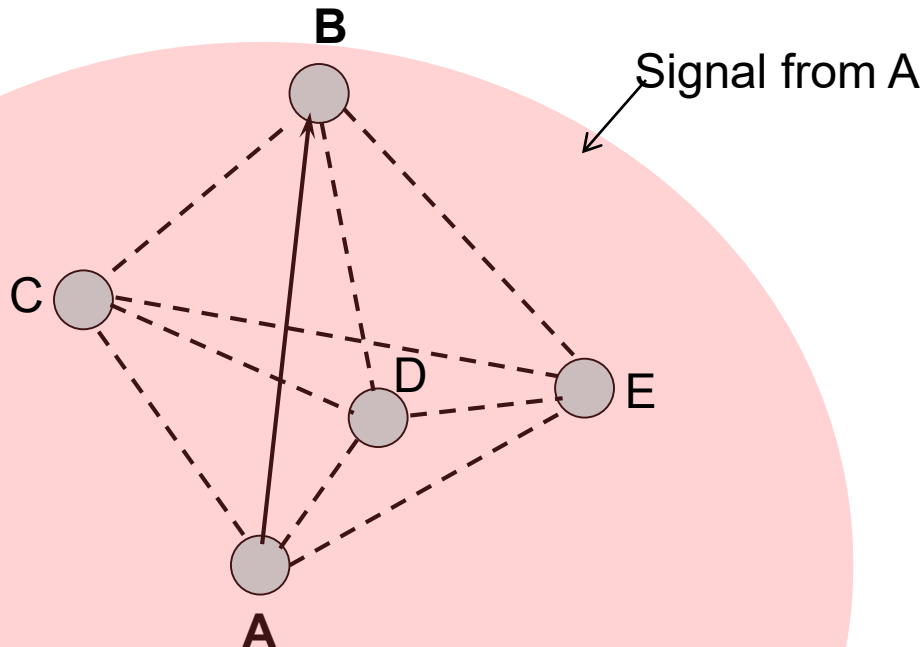
- Performance degradation in large networks
- Near-far problem



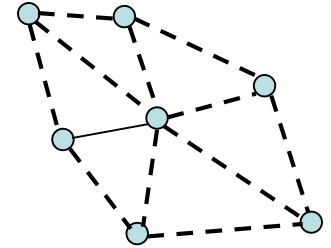
Fully Connected peer-to-peer

- Disadvantages

- Transmitters operating at high power levels (to reach a far station) will interfere with unintended receivers in close proximity



Multihop Peer-to-Peer



- Advantages

- Transmit power efficiency
- Only solution if no infrastructure available
- Widely used in military applications
- Gaining popularity in other types of wireless networks
 - Ad hoc networks
 - Sensor networks

- Disadvantages

- Complex algorithms for efficient message routing and control
- Multiple store-and-forwards increase delay for users separated by multiple hops
- Overhead to set up efficient routes
- No central timing or power control authority



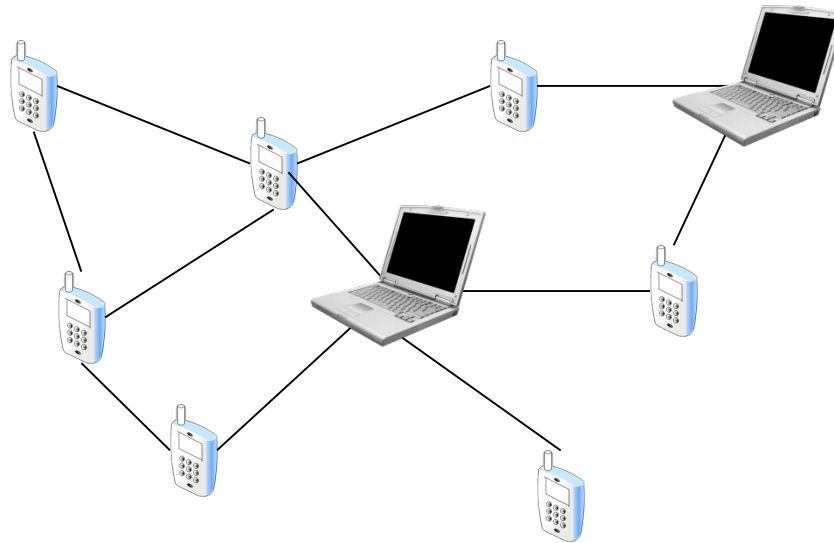
Types of Wireless Networks

- **Mobile Ad hoc Networks**
 - Multi-hop peer-to-peer networks
 - Hybrid networks
 - Unicast, multicast and broadcast networks
- **Wireless Sensor Networks**
 - Different applications uses different network topologies
 - Converge-cast, unicast, broadcast and multicast networks
- **Wireless Mesh Networks**
 - Similar to Ad Hoc, but more static/permanent, with little or no mobility of router nodes
 - Less resource constraints compared to Ad Hoc



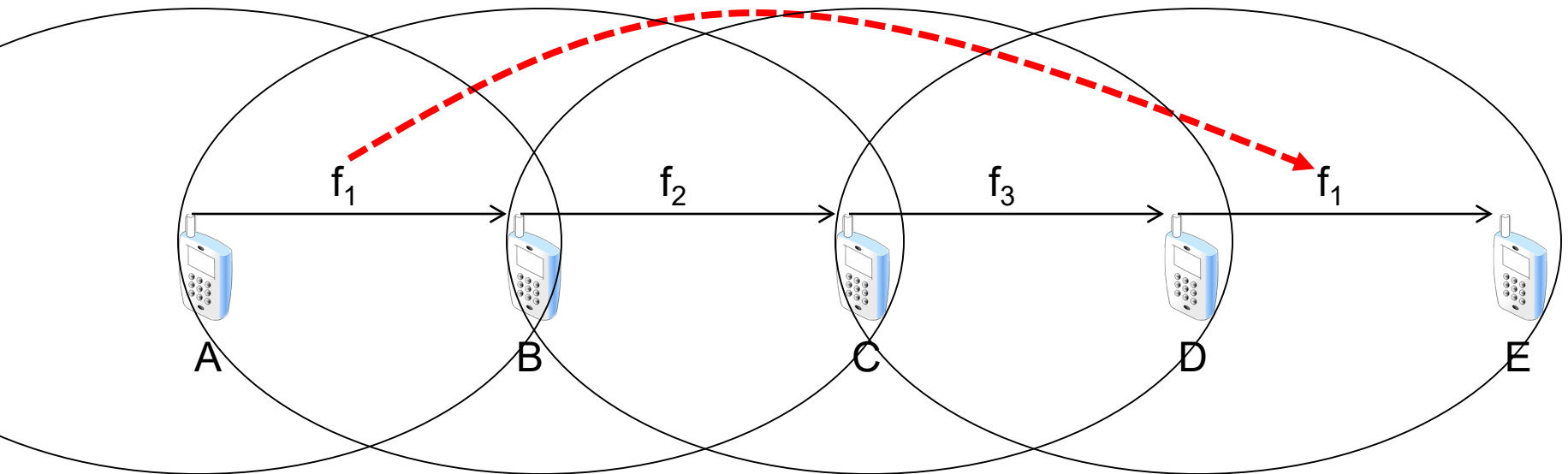
Mobile Ad Hoc Networks (MANETs)

- Definition
 - A collection of wireless mobile hosts and IoT devices forming a temporary network without the aid of any centralized administration or standard support IoT services



MANETs

- Mobile nodes have limited communication range
 - Reduces battery drain
 - Enables spatial reuse of limited bandwidth → increased network capacity



MANETs

- To connect all nodes in the network, each node is a
 - Packet source
 - Packet sink
 - Router
- Nodes must route packets for other nodes to keep the network fully connected
- In MANETs, a big problem is how to determine where a destination node is located relative to a sending node



MANETs

- Route-finding is a critical issue
 - Want to determine an “optimal” way to find “optimal” routes
- Dynamic links
 - Broken links must be updated when a node moves out of communication range with another node
 - New links must be formed when a node moves into communication range with another node
 - Based on this new information, routes must be modified
- Frequency of route changes a function of node mobility



Issues in MANETs

- Routing performance
 - Routes change over time due to node mobility
 - Would like to avoid long delays when sending packets
 - But would like to avoid lots of route maintenance overhead
 - Want as many participating nodes as possible for greater aggregate throughput, shorter paths, and smaller chance of partition
- MAC (medium access control)
 - Broadcast communication channel
 - Neighbor nodes change over time
 - Nodes sleep to reduce energy drain
 - No coordination/cooperation among nodes



Issues in MANETs

- Quality of service
 - Link variability → route variability
 - Collisions
 - Congestion
- Security -interesting new vulnerabilities and complexities
 - Routing denial of service
 - Nodes may agree to route packets
 - Nodes may then fail to do so
 - Broken, malicious, selfish nodes
 - Key distribution and trust issues



Wireless Sensor Networks (WSNs)

- Networks of Sensor nodes communicating wirelessly
- Sensors of wide variety monitor environment
 - Cameras, microphones, physiological sensors, etc.
 - Gather data for some purpose
- Microsensors
 - Low power, cheap sensors
 - Sensor module (e.g., acoustic, seismic, image)
 - Comes with a digital processor for signal processing and network protocol functions
 - Uses Radio for communication
 - Battery-operated
- Microsensor data limited in range and accuracy
 - Each node can only gather data from a limited physical area
 - Data may be noisy
 - Data aggregation enables higher quality (less noisy) data, gives information about larger physical area than individual data



Wireless Sensor Networks (WSNs)

- Hundreds or thousands of nodes scattered throughout an environment
- New wireless networking paradigm
 - Requires autonomous operation
 - Highly dynamic environments
 - Sensor nodes can fail/be added
 - Events in the environment can affect operation
 - Distributed computation and communication protocols required



Wireless Mesh Networks (WMN)

- Multihop decentralized Networks Similar to Ad Hoc Networks

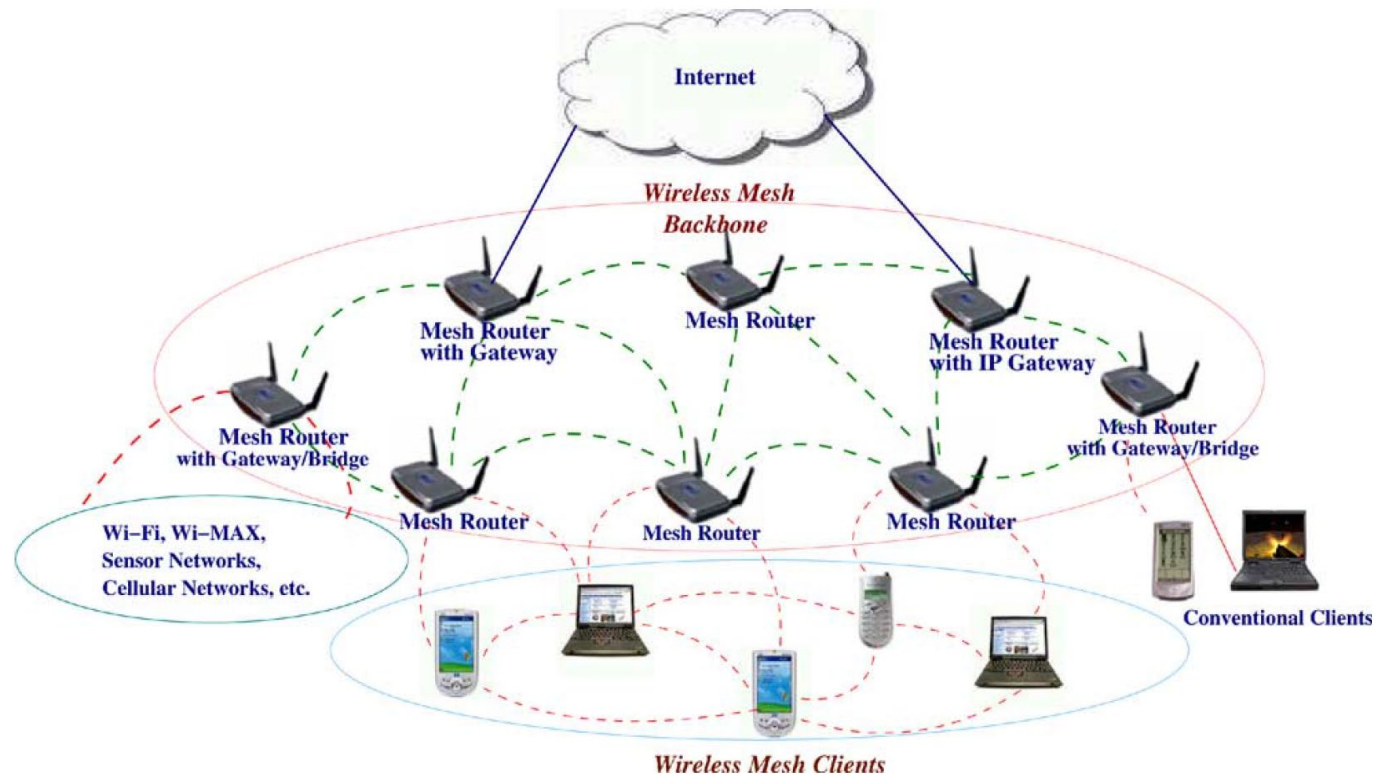


Fig. Source: Akyildiz et al, "Wireless mesh networks: a survey", Computer Networks



WMNs

- Consists of routing nodes which are usually fixed, and mobile users
- Energy constraint is less of an issue compared to MANETs
- Processing power is more accessible
- Critical resource is bandwidth
- Scalability is another important issue (A network is scalable when its performance does not degrade as the network size increases)
- Security is still a major issue



MEDIUM ACCESS CONTROL (MAC)



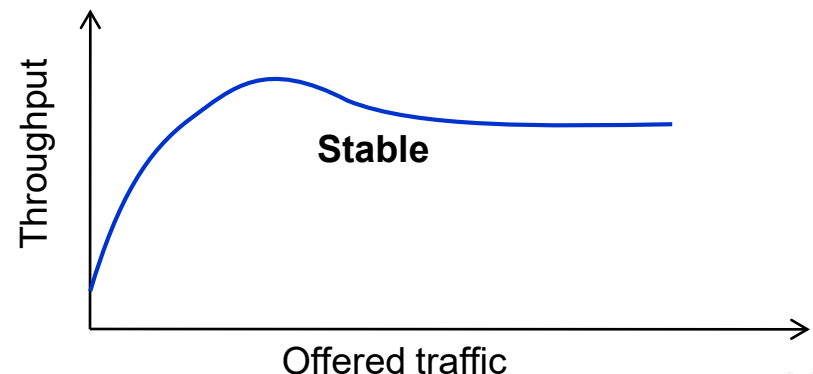
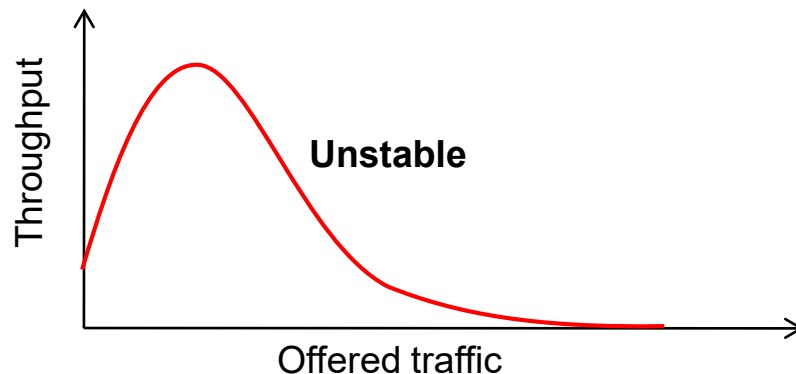
What is Medium Access Control?

- Mechanisms (protocols) that enable multiple users to share a finite amount of frequency and time resources
 - Needed for efficient operation and good performance for wireless systems
 - Goal: minimize overhead while maximizing overall network capacity



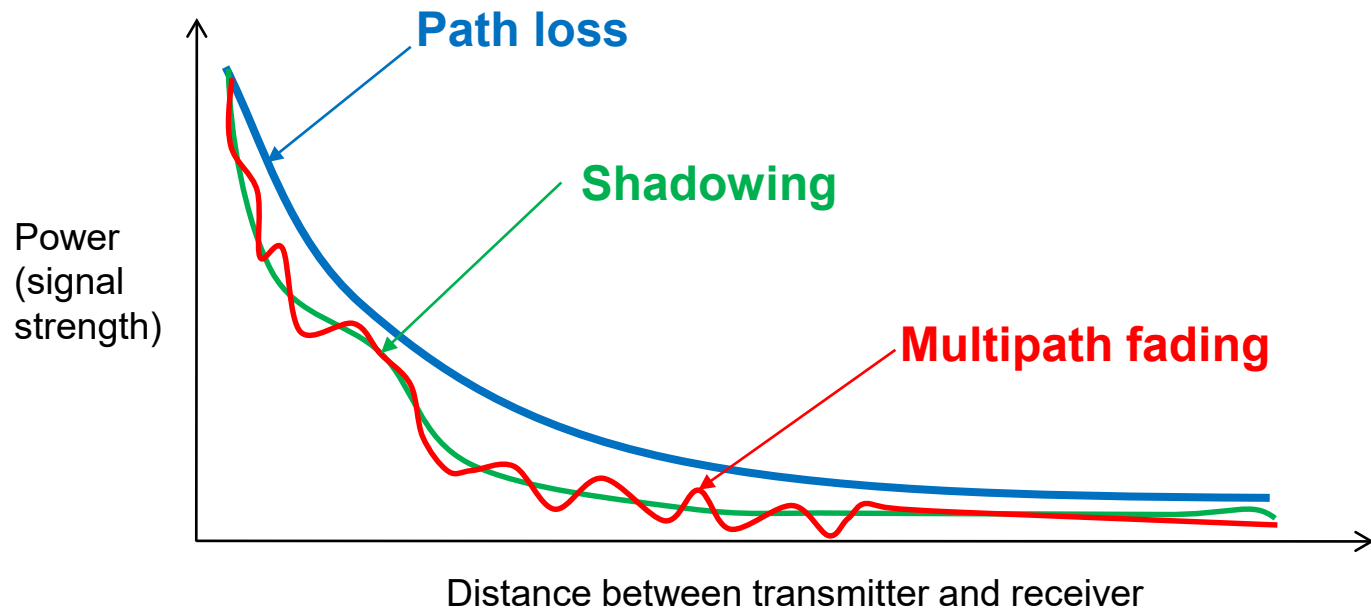
Performance Metrics of MAC Protocols

- Throughput (S): average number of messages successfully transmitted per unit time
- Delay (D): average delay experienced by a packet
- Fairness: how well a MAC protocol shares the bandwidth among multiple users
- Stability: performance under load fluctuations



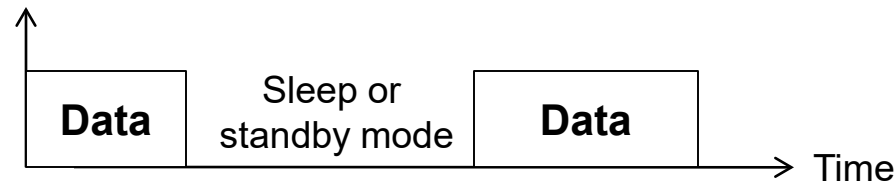
Performance Metrics of MAC Protocols

- Robustness against channel fades

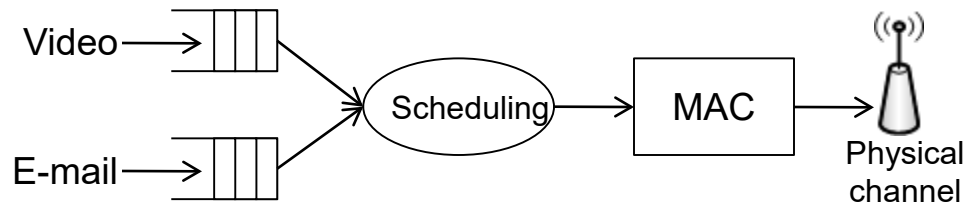


Performance Metrics of MAC Protocols

- Power consumption: power-saving features of the protocol

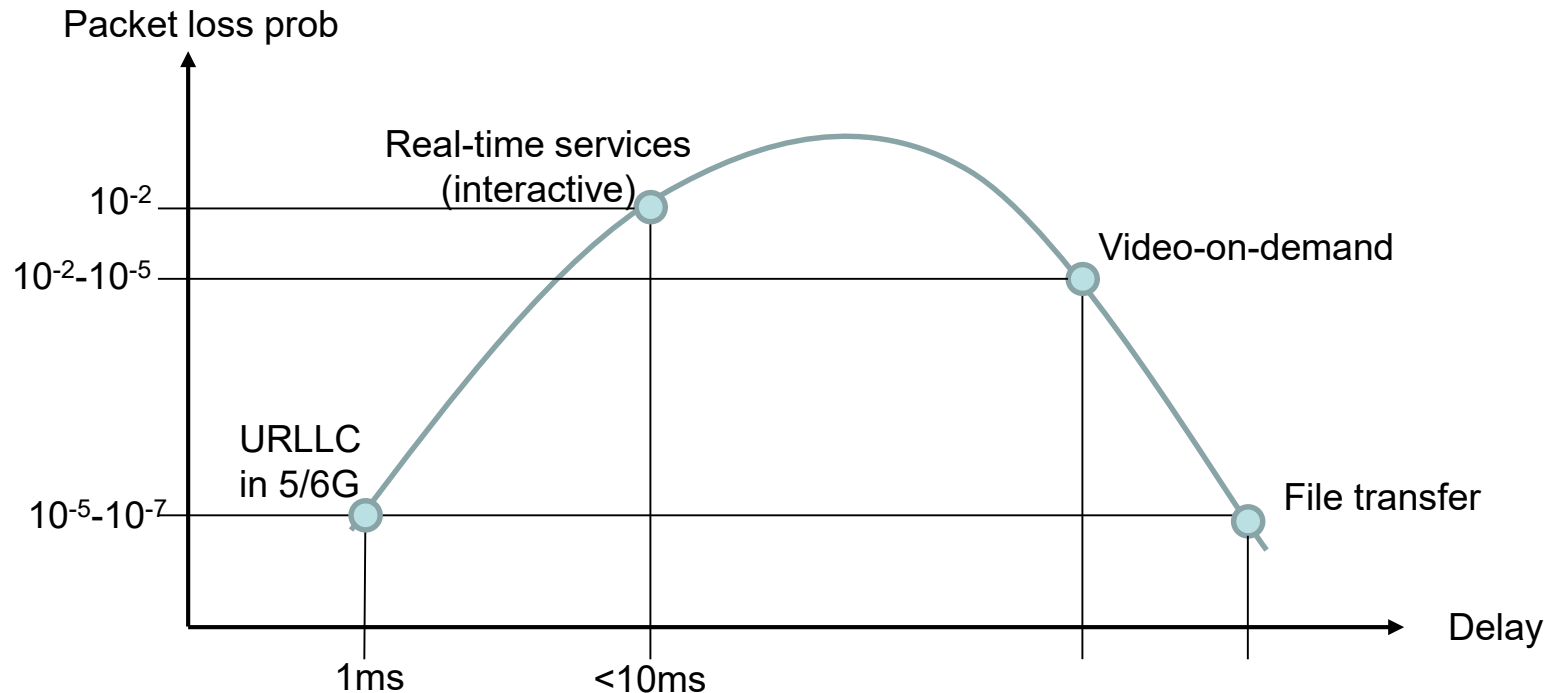


- ◆ Multimedia support: how well the protocol supports different types of traffic (e.g., real-time, high-priority data, etc.)



Performance Metrics of MAC Protocols

- Packet loss probability versus delay



URLLC: Ultra-Reliable and Low-Latency Communications

C. She, C. Sun, Y. Li, C. Yang, H. V. Poor and B. Vucetic, "Toward Ultra-Reliable and Low-Latency Communications in 6G: From Model-Based Approaches to Deep Learning,"

P. Dymarski, S. Kula, and T. N. Huy, "QoS conditions for VoIP and VoD," J. Telecommun. and Inf. Tech., pp. 29–37, Mar. 2011.



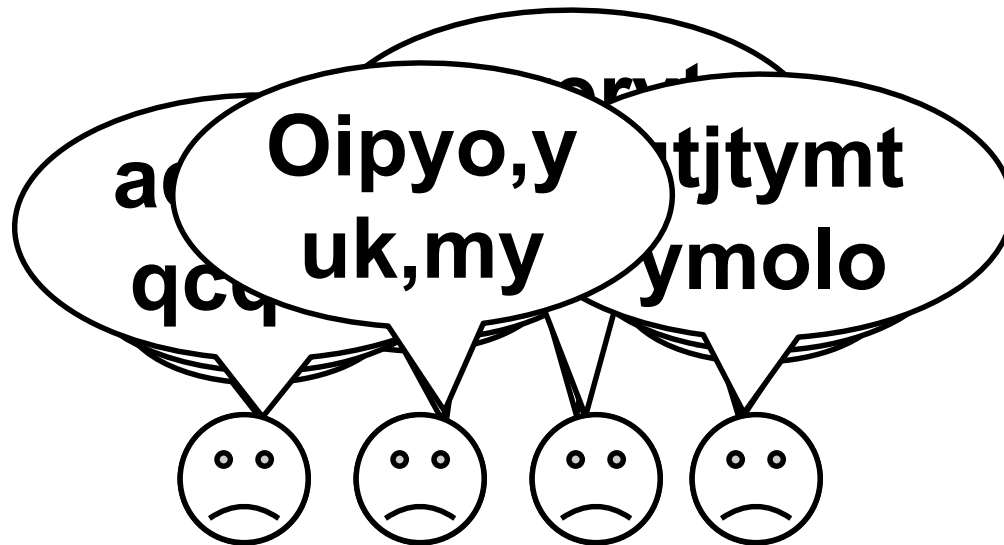
Main Types of MAC

- Fixed assignment techniques
 - Frequency-division multiple access (FDMA)
 - Time-division multiple access (TDMA)
 - Spread-spectrum multiple access (SSMA)
 - Direct-sequence spread-spectrum (DSSS) / Code-division multiple access (CDMA)
 - Frequency-hopped spread-spectrum (FHSS)
 - Space-division multiple access (SDMA)
- Random Access (RA) techniques
 - ALOHA
- Controlled random access (hybrid)



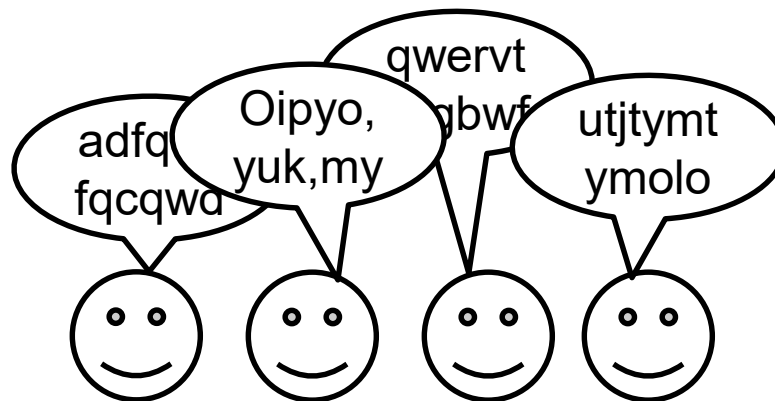
Analogy from Human Communication

- Suppose there are people at a party who all want to talk
- If everyone talked at once, no one would be able to hear anyone
- If one person raises her/his voice to be heard, others will raise their voices and eventually everyone will be shouting and no one will be able to communicate



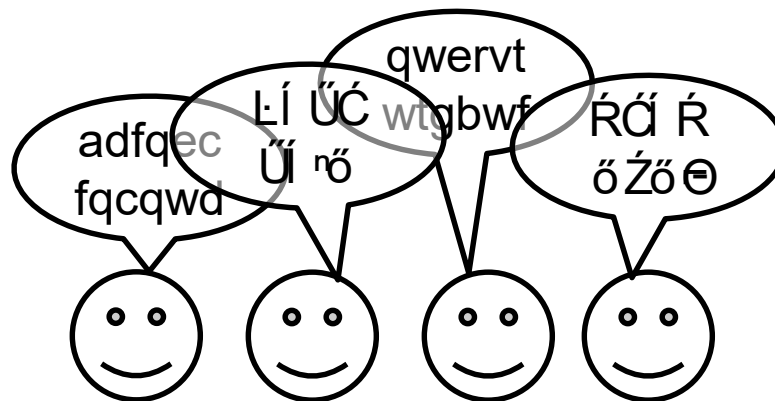
Analogy from Human Communication

- How can this situation be resolved?
 - Each person could be given a turn to speak (TDMA)



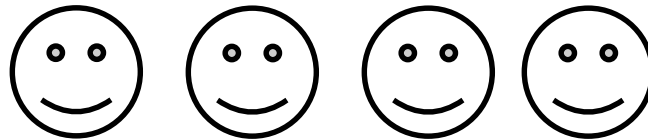
Analogy from Human Communication

- How can this situation be resolved?
 - Each group could be given a language in which to speak to each other (CDMA)



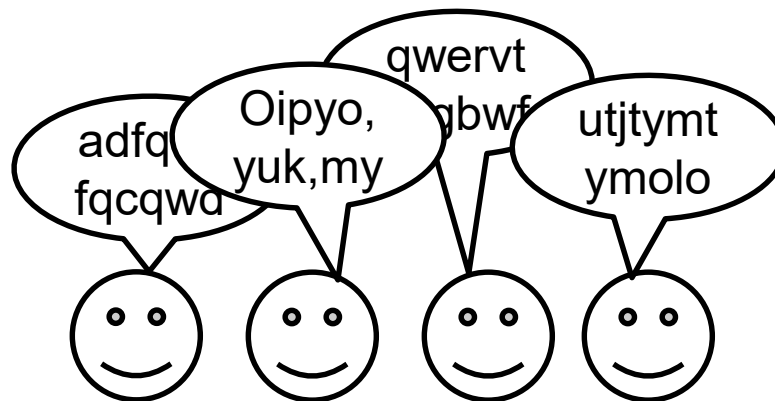
Analogy from Human Communication

- How can this situation be resolved?
 - Each group could be given a corner of the room to hold their conversation (SDMA)



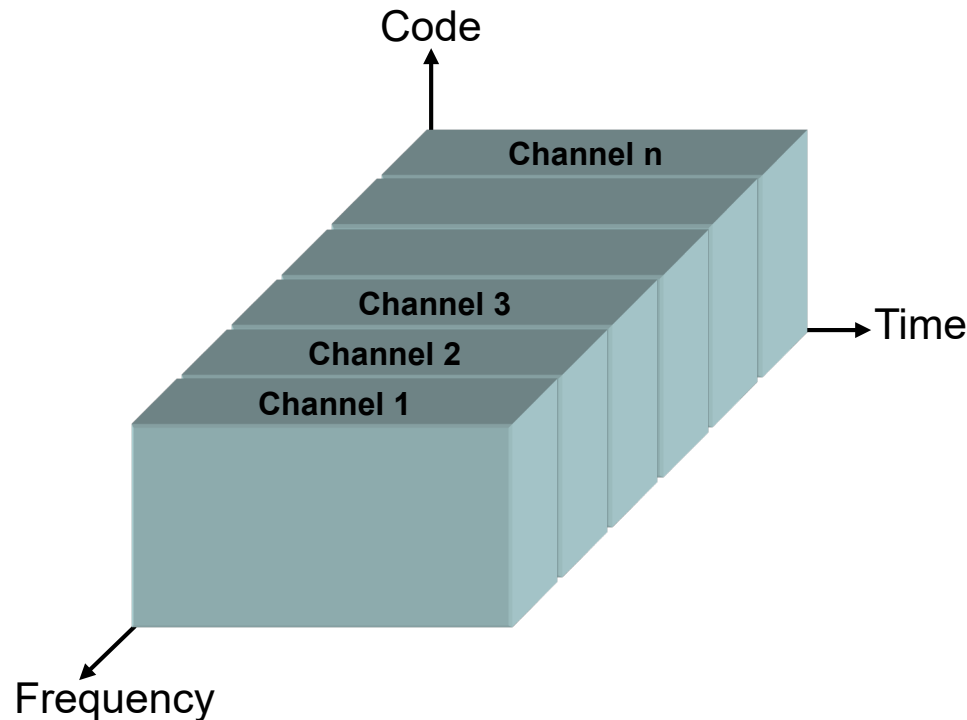
Analogy from Human Communication

- How can this situation be resolved?
 - Each person talks with a certain probability p when they determine that no one else is currently speaking (PR)



Frequency Division Multiple Access (FDMA)

- Radio spectrum broken into frequency bands (channels)
- Each channel allocated to a different user (only 1 user per frequency band)
- Each channel must contain guard bands



FDMA

- Channels can be assigned on-demand when a user needs to communicate
- Each user can only be assigned 1 channel → if not enough users for the number of channels, the radio spectrum is unused (i.e., wasted)
- FDMA is usually used in narrowband systems (e.g., 30 kHz frequency bands)
- Little synchronization is required because transmission is continuous in FDMA → reduces overhead



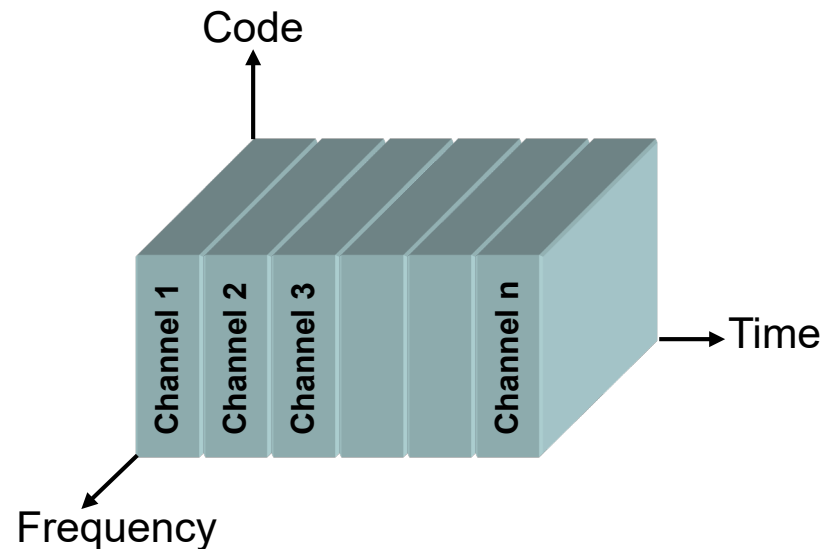
FDMA

- Requires expensive filters to reduce adjacent channel interference
- Intermodulation(IM)
 - Nonlinearities in power amplifiers cause signal spreading in freq domain
 - Undesired RF radiation that leaks into other channels in FDMA systems
 - Adjacent-channel interference
 - Generation of undesirable harmonics that cause interference to other users in mobile system or other systems in adjacent spectrum bands



Time Division MA (TDMA)

- Users access entire radio spectrum for a given time slot (channels)

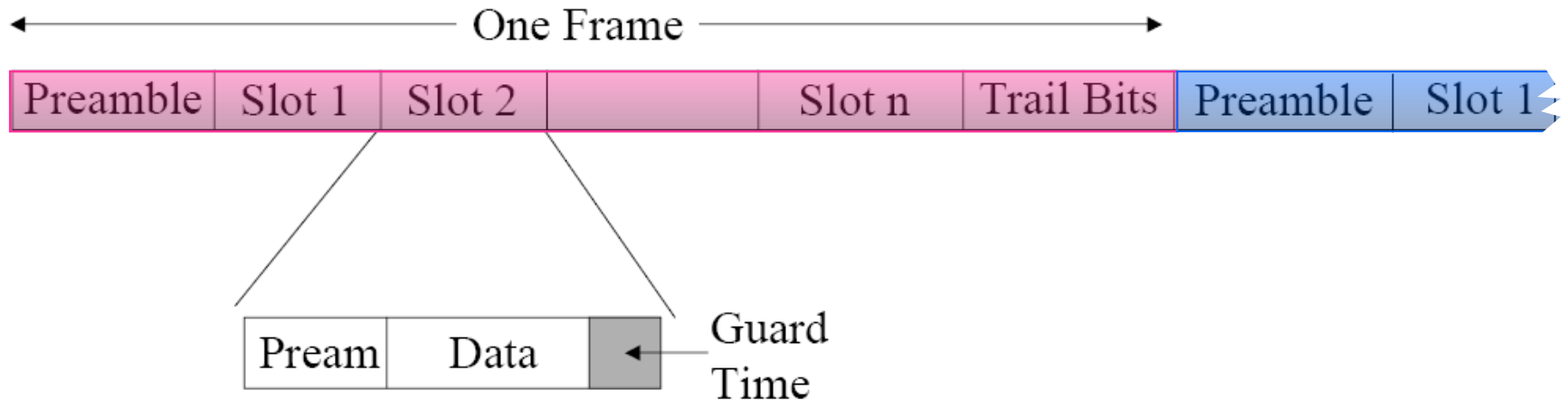


- ◆ Only 1 user can transmit or receive data per slot
- ◆ Time divided into frames
 - Preamble (with synchronization info)
 - Several slots of data
 - Number of data slots per frame depends on modulation, bandwidth, average data rates and required latencies



TDMA (cont.)

- Slot contains
 - Preamble for addressing and synchronization
 - Data
 - Guard times between the slots to reduce cross-talk between channels



TDMA (cont.)

- Can use multi-slot assignments to support increased user data rate
- Channels can be assigned on-demand when a user needs to communicate

TDMA Advantages

- Narrowband filters not needed → cheaper than FDMA
- Mobile devices can save battery power by turning off transmitter and receiver during slots when not transmitting or receiving data
- Can allocate multiple time slots to a user to provide increased data rate → more BW efficient than FDMA

TDMA Disadvantages

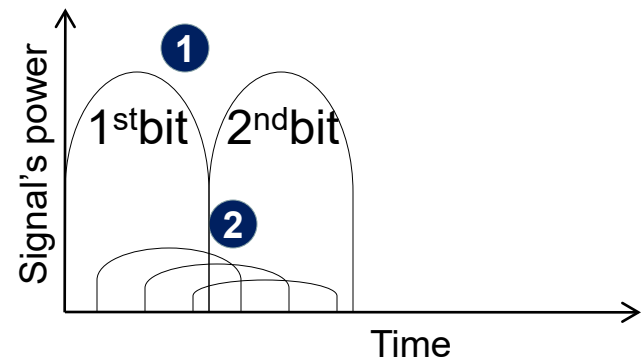
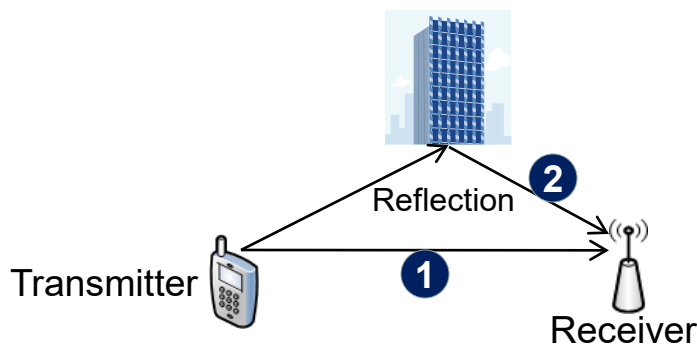
- Requires guard time between time slots to separate users and accommodate
 - Time inaccuracies due to clock instability
 - Delay spread of transmitted symbols
 - Transmission time delay
- Requires signal processing techniques and high overhead for synchronization due to burst transmissions

Spread-Spectrum MA (SSMA)

- Transmission bandwidth is several orders of magnitude greater than minimum required bandwidth
- Multiple users share same RF spectrum at same time
 - No hard limit on the number of users
 - Bandwidth inefficient when few users sharing the channel
 - Bandwidth efficient when lots of users share the channel
- Direct-sequence (DS): pseudo-noise (PN) sequence converts narrowband signal to wideband signal with noise-like properties
- Frequency-hopped (FH) spread spectrum: pseudo-random hopping sequence used to vary carrier signal

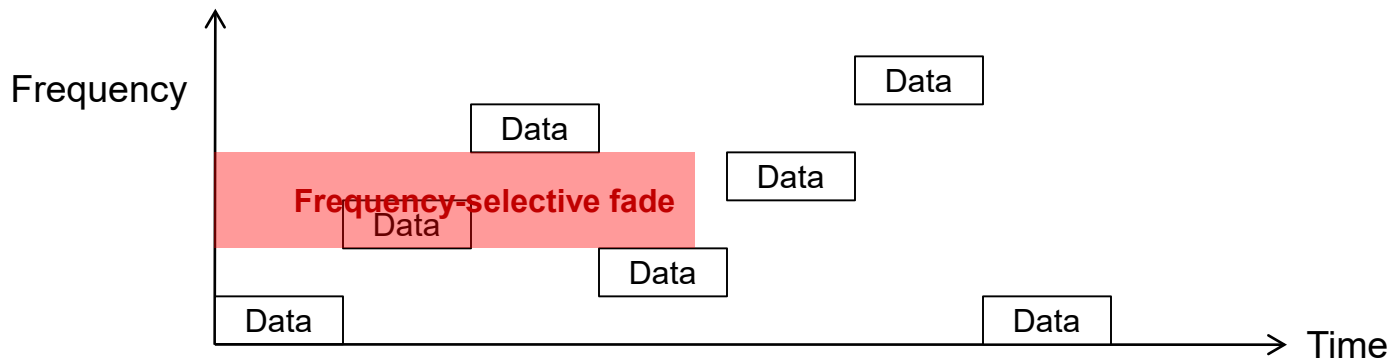
SSMA Advantages

- SS signals can be overlaid onto bands where other systems are already operating with minimal impact to or from the other systems
- Unlicensed SS systems can be used in the ISM bands
- SS signals have good immunity to mutli-path



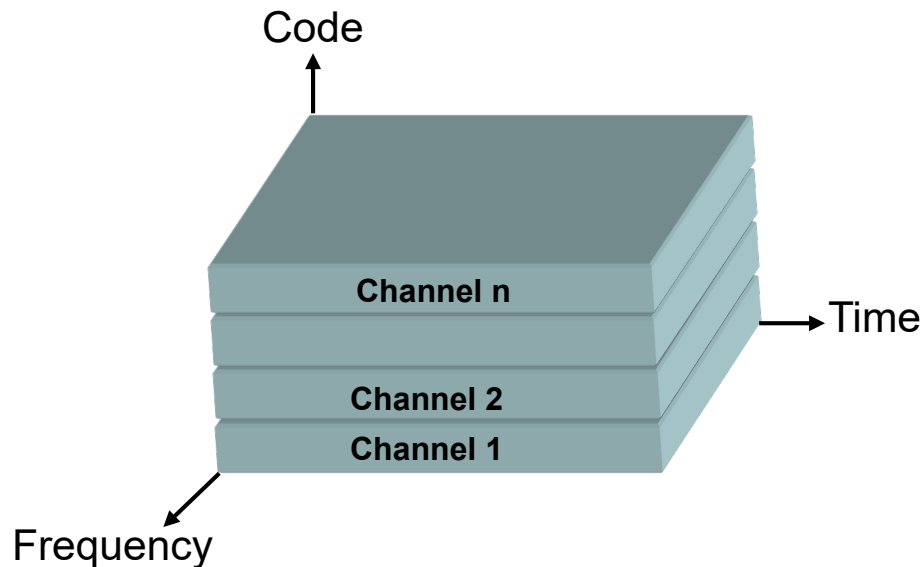
SSMA Advantages

- SS signals have good immunity to frequency-selective fading

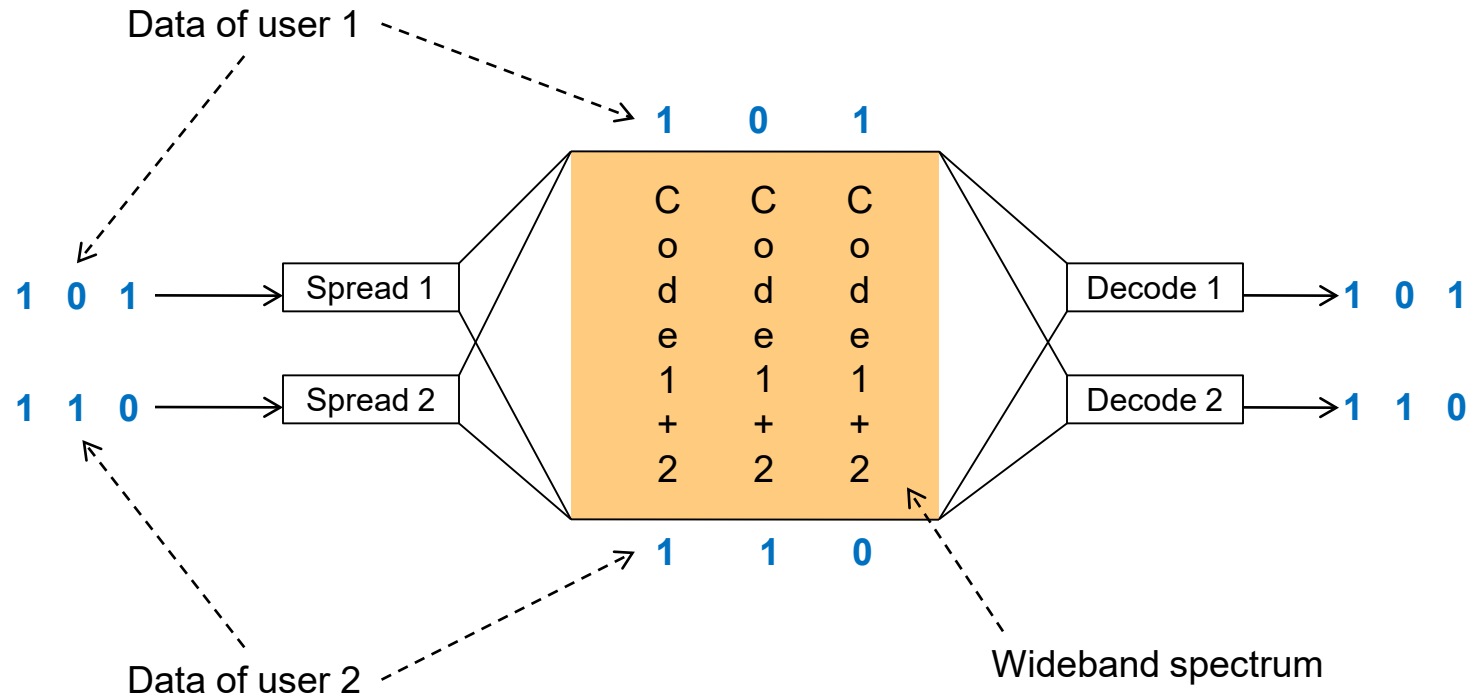


Direct-Sequence SS (DSSS)

- Each user occupies bandwidth several times larger than the message bandwidth
- Message signal multiplied by a very large BW “spreading signal”, a PN-code sequence with a “chip rate” several orders of magnitude larger than the message data rate
- In CDMA (code-division multiple access), spreading signals (codes) are approximately orthogonal



Direct-Sequence SS (DSSS)



DSSS (cont.)

- Codes' orthogonality
- with code $c_i(t)$ will filter out other orthogonal signals since

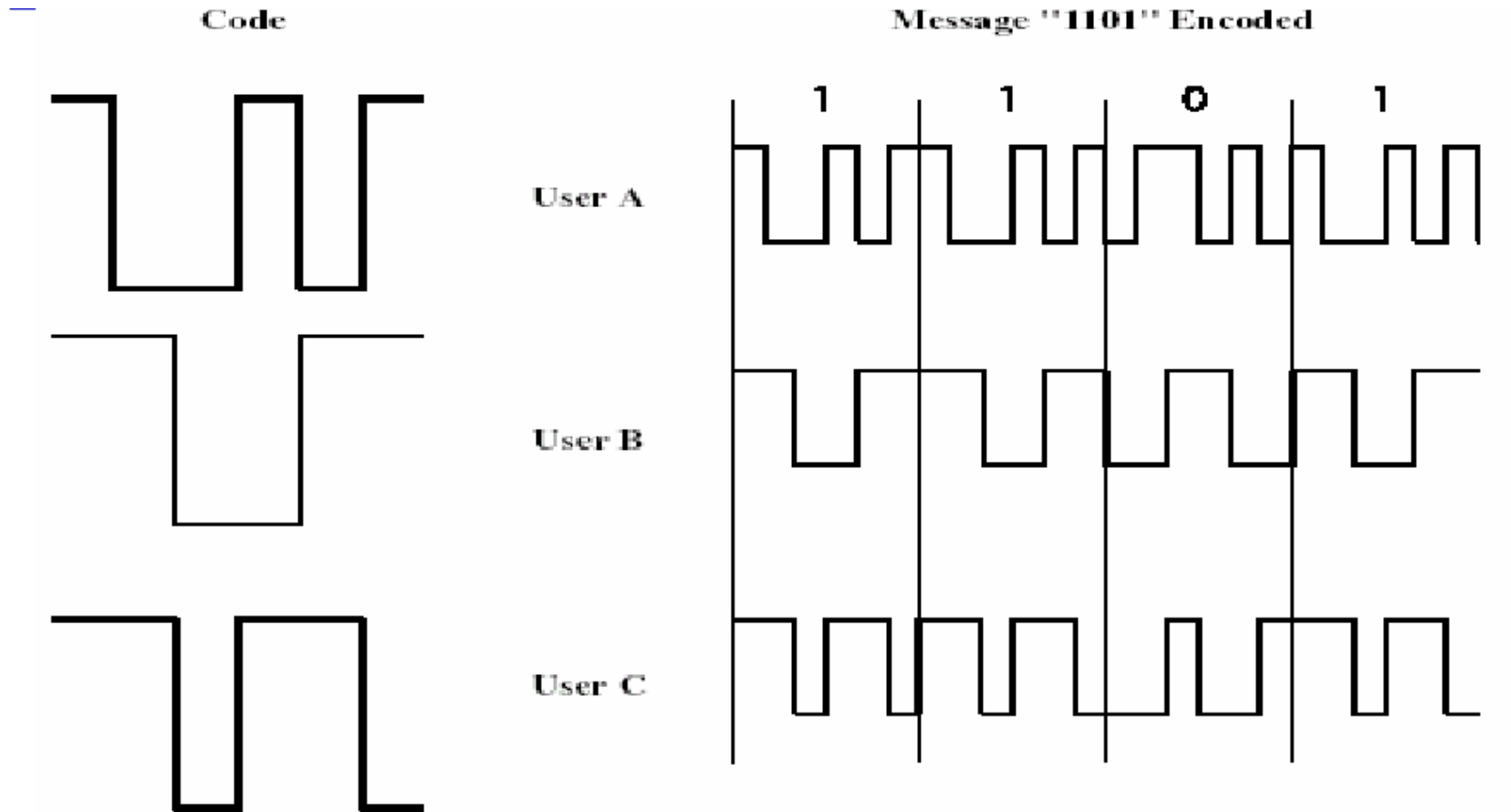
$$\int_{-\infty}^{\infty} c_i(t) \cdot c_j(t) dt = 0$$

DSSS Example

- User A code = $\langle 1, -1, -1, 1, -1, 1 \rangle$
 - For a 1 bit, A sends $\rightarrow \langle 1, -1, -1, 1, -1, 1 \rangle$
 - For a 0 bit, A sends $\rightarrow \langle -1, 1, 1, -1, 1, -1 \rangle$
- User B code = $\langle 1, 1, -1, -1, 1, 1 \rangle$
 - For a 1 bit, B sends $\rightarrow \langle 1, 1, -1, -1, 1, 1 \rangle$
 - For a 0 bit, B sends $\rightarrow \langle -1, -1, 1, 1, -1, -1 \rangle$
- Check that A and B are orthogonal $\rightarrow$ Dot product should be zero.

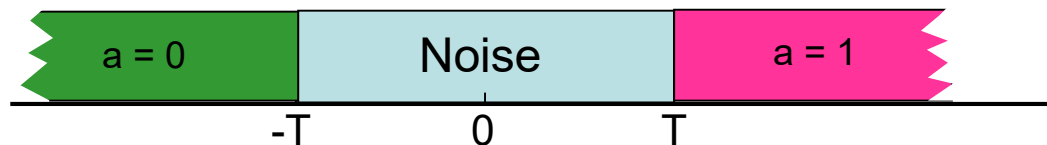
$$\begin{aligned}\langle A.B \rangle &= 1.1 + (-1).1 + (-1).(-1) + 1.(-1) + (-1).1 + 1.1 \\ &= 1 - 1 + 1 - 1 - 1 + 1 = 0\end{aligned}$$

DSSS Example



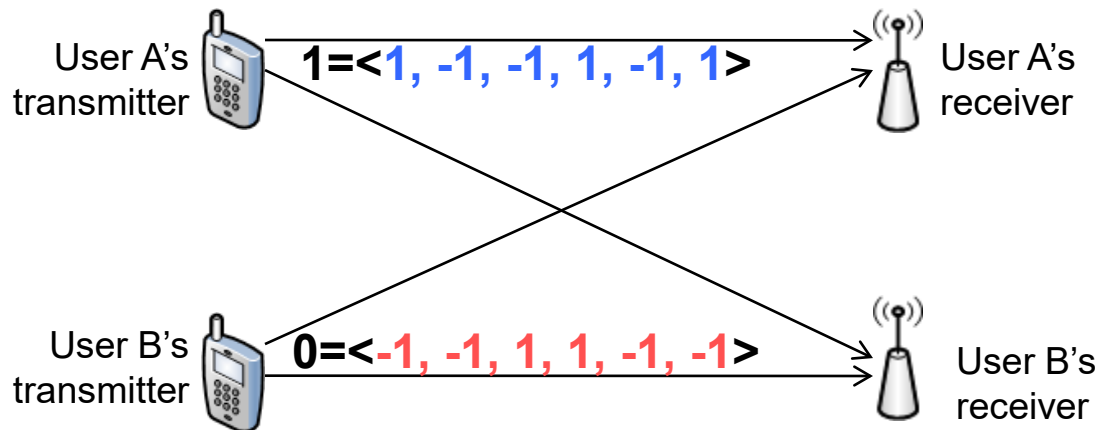
DSSS Example

- Receiver receives A's information by
 - $S = (\text{A's code}) \times (\text{received chip pattern})$
- Receiver receives B's information by
 - $S = (\text{B's code}) \times (\text{received chip pattern})$
- Decision is based on a threshold T ():
 - output a binary 0 if $S < -T$
 - output a binary 1 if $S > T$
 - treat as noise if $-T \leq S \leq T$



DSSS Example

- Let's assume user A sends a '1' bit and user B sends a '0' bit to a base station. Let's see how the base station tells A's information from B's information. Suppose Threshold $T = 4$.



DSSS Example

- Let's assume user A sends a '1' bit and user B sends a '0' bit to a base station. Let's see how the base station tells A's information from B's information. Suppose Threshold $T = 4$.

Signal from transmitter A	1	-1	-1	1	-1	1		Rx A interprets this as '1' (Since 6 >4)
A's code	1	-1	-1	1	-1	1		
Multiplication	1	1	1	1	1	1	=6	
Signal from transmitter B	-1	-1	1	1	-1	-1		Rx A interprets this as 'Noise' (Since -4 < 0 < 4)
A's code	1	-1	-1	1	-1	1		
Multiplication	-1	1	-1	1	1	-1	=0	
T							=4	

Interpretation of A's code for signal from transmitters A and B

DSSS Example

Signal from transmitter A	1	-1	-1	1	-1	1	
B's code	1	1	-1	-1	1	1	
Multiplication	1	-1	1	-1	-1	1	= 0
Signal from transmitter B	-1	-1	1	1	-1	-1	
B's code	1	1	-1	-1	1	1	
Multiplication	-1	-1	-1	-1	-1	-1	= - 6
T							= 4

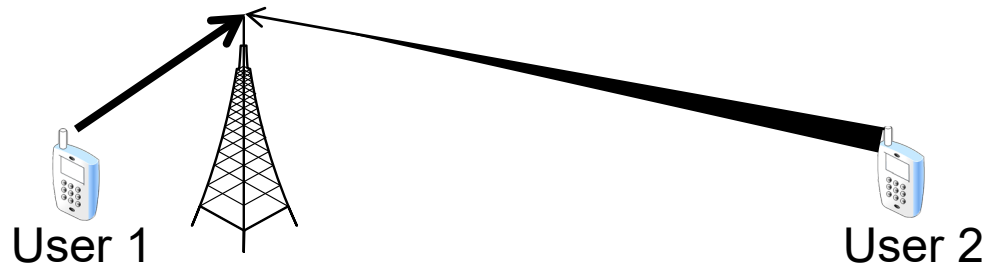
Rx B
interprets
this as
'Noise'
(Since
 $-4 < 0 < 4$)

Rx B
interprets
this as '0'
(Since
 $-6 < -4$)

Interpretation of B's code for signal from transmitters A and B

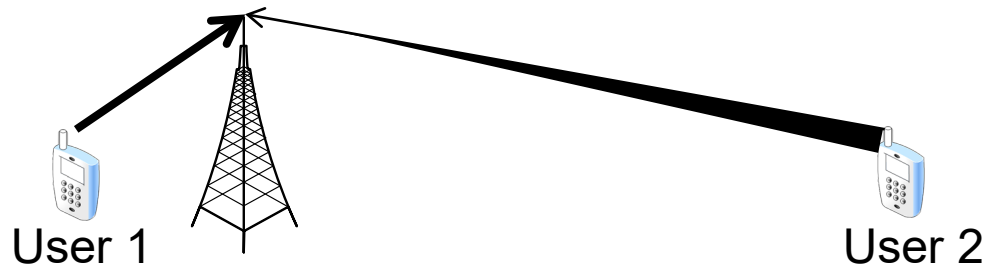
Near-Far Problem

- If all mobiles transmit with the same power, the signals from mobiles closest to the BS will be received with much larger power than signals from mobiles further away. Therefore, the SNR for signals from mobiles far from the BS will be low
 - Signals from mobiles close to the BS will drown out signals from mobiles far away from the BS



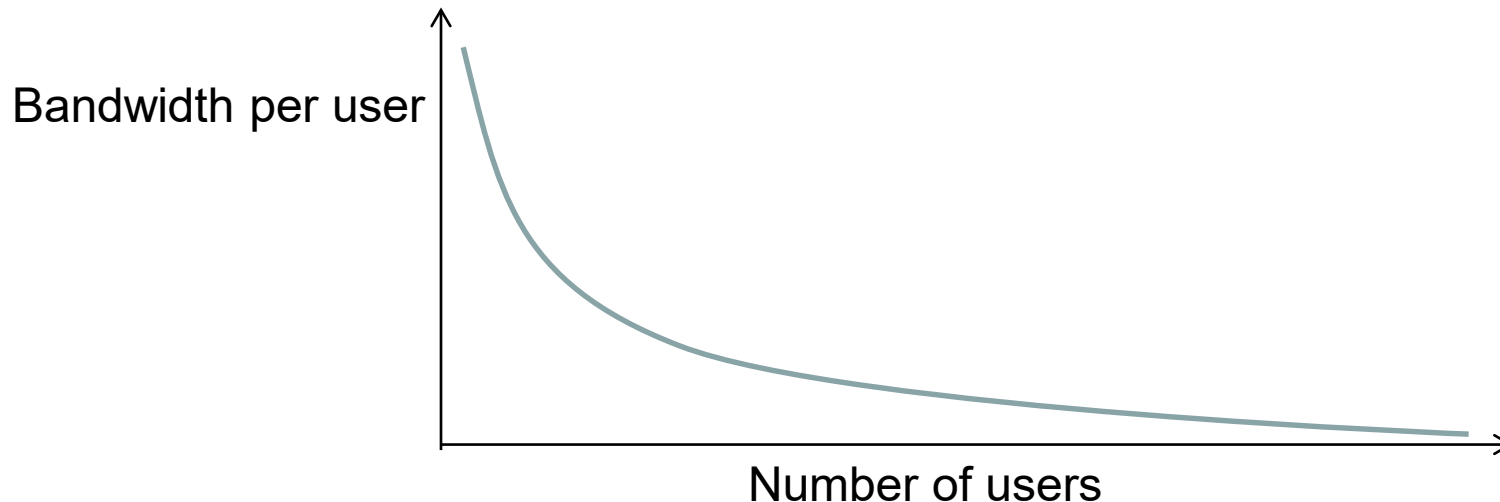
Near-Far Problem

- Requires power control to ensure the power of all signals received at the base station is approximately equal
 - BS determines received power from each mobile (using RSSI—**R**eceived **S**ignal **S**trength **I**ndicator)
 - Tells the mobile to increase or decrease its transmit power to ensure all signals received with the same power
- Power control conserves battery power and minimizes interference to other users



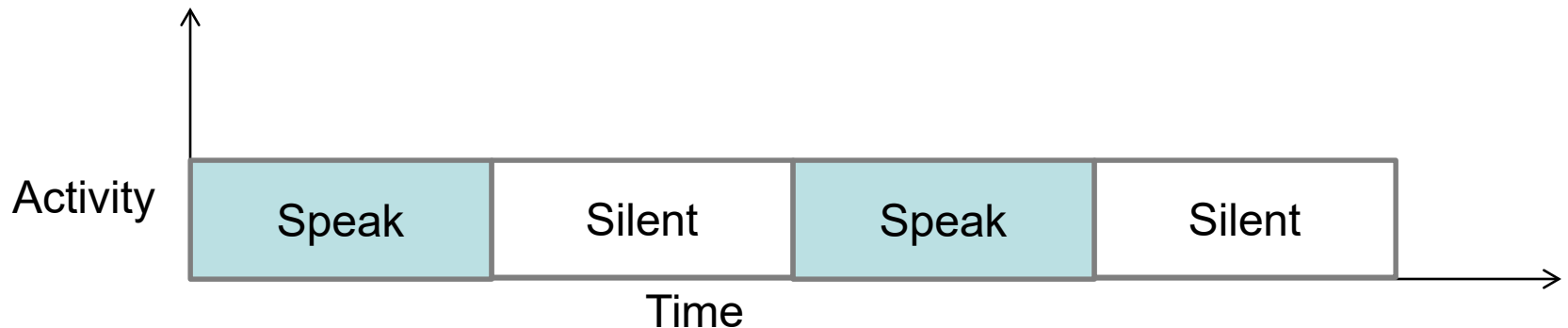
Advantages of DSSS

- Multipath fading and frequency-selective fading effects reduced due to spreading of the signal bandwidth
- **Soft capacity limit** on the number of users—each user sees degradation in performance as # of users increased and improvement in performance as # of users decreased



Advantages of DSSS

- Can take advantage of voice activity patterns that cannot be effectively exploited in FDMA or TDMA systems
- Security: need to know spreading code to intercept signal



Disadvantages of DSSS

- Implementation complexity: high channel data rates
 - Expensive receivers
- Self-jamming due to non-perfect orthogonality of spreading codes
 - Other users' signals will appear as noise and reduce SNR of desired signal
- Near-far problem causes reduced performance if receiver cannot control the power level of the near mobile

Frequency-Hopped SS (FHSS)

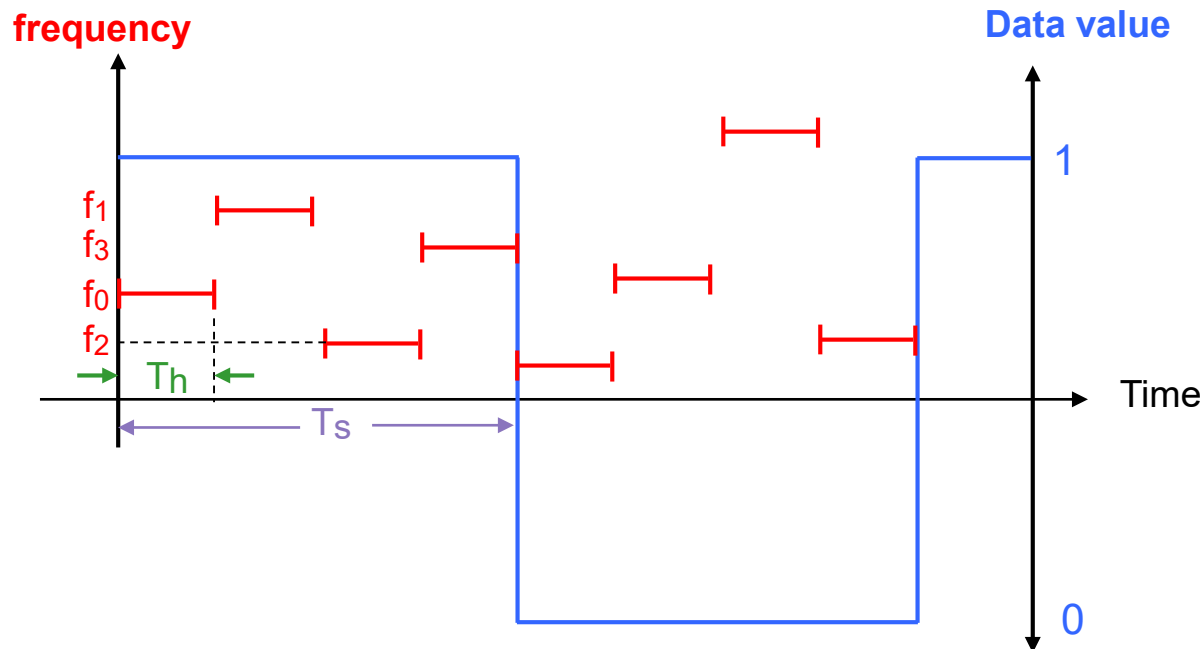
- System bandwidth (B_t) broken into smaller channels (bandwidth = B_c), data stream broken into bursts
- Carrier frequency varied for each data burst in a pseudorandom fashion
 - User “hops” among the different channels
 - T_h = hop period
 - Instantaneous bandwidth of each transmission is small, but spread bandwidth is large
- Multiple users can access the channel at the same time due to pseudorandom hopping sequence
 - Ensures low probability of multiple users accessing same hop frequency at the same time

FHSS

The rate of Frequency Hopping can be **faster** or **slower** than the symbol rate (the rate at which the data value changes)

- **Fast frequency hopping**

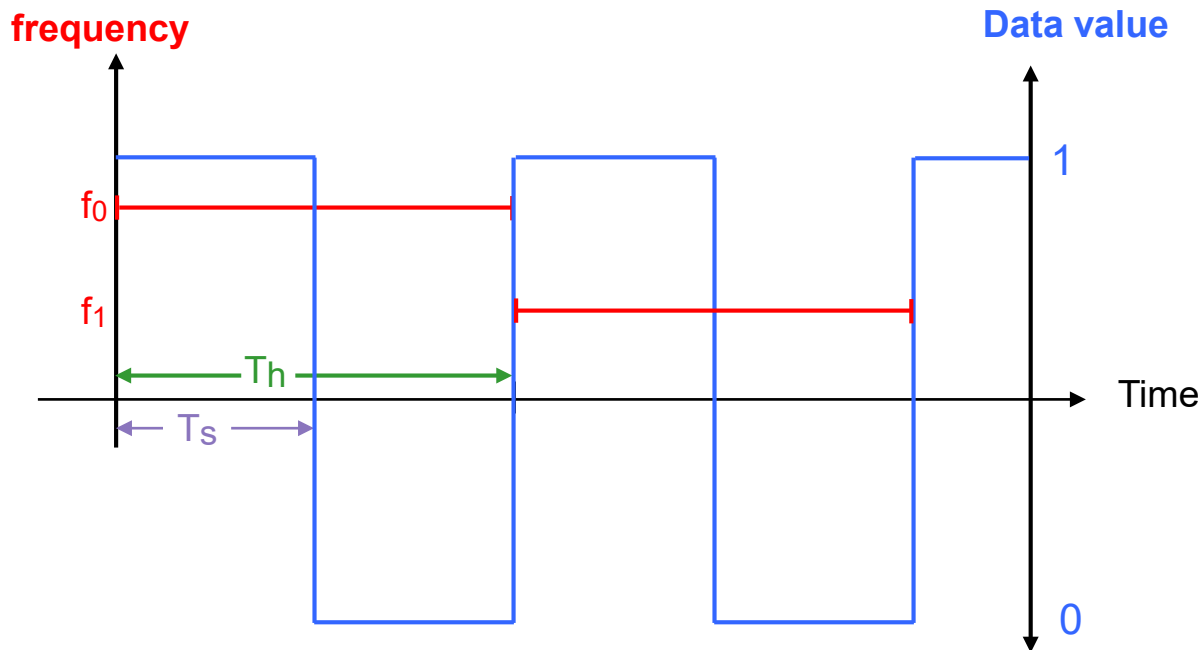
- $T_h < T_s \rightarrow$ there is a frequency hop for each transmitted symbol
- Hopping rate equals or exceeds the information symbol rate



FHSS

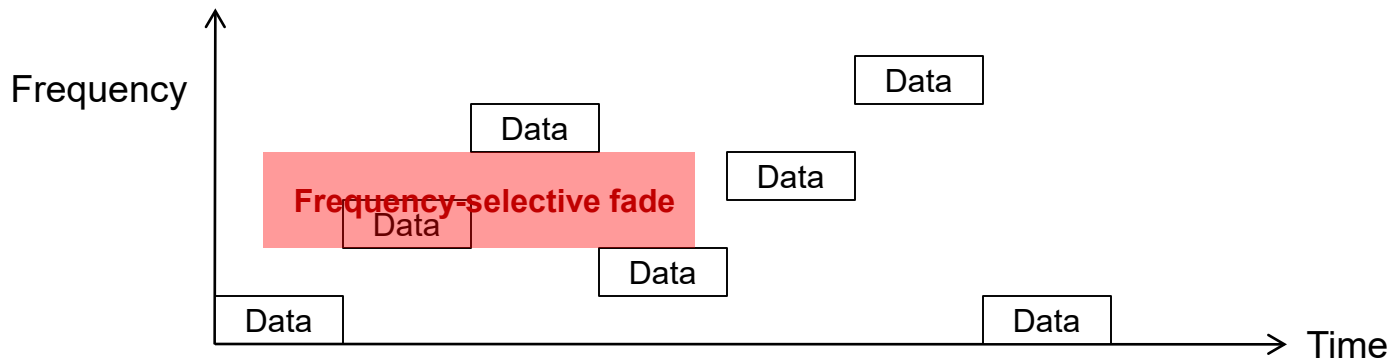
- **Slow frequency hopping**

- $T_h \geq T_s \rightarrow$ one or more symbols are transmitted in the time interval between frequency hops



Advantages of FHSS

- Immunity to frequency-selective fades since the transmission hops over different channels
- Security: use of pseudorandom hopping pattern makes it hard for others to intercept full message
- Soft capacity
- Immune to the near-far problem

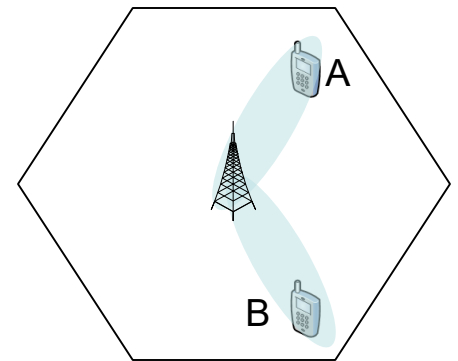


Disadvantages of FHSS

- Multi-user interference possible if two or more users simultaneously occupy a frequency channel

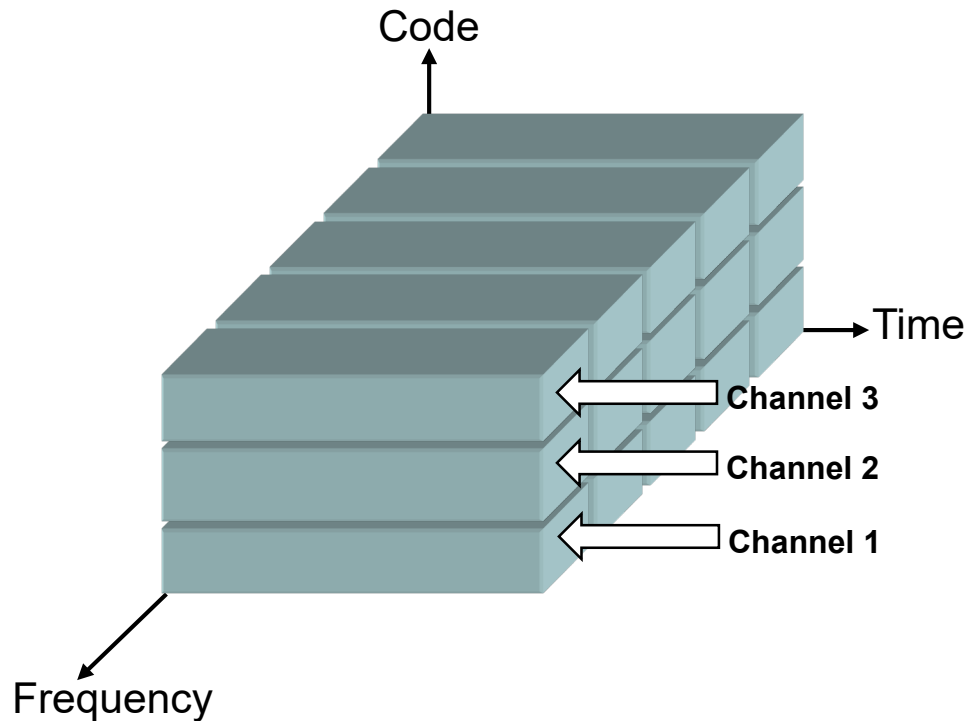
Space-Division MA (SDMA)

- Divides a coverage area in space into segments
- Uses directional antennas that transmit signals only in a certain direction
 - Within each space sector, can use TDMA, FDMA, or SSMA to divide the time-frequency resources among users
 - Adaptive antennas track users and direct energy in direction



Hybrid MA

- FDMA/CDMA
 - Spectrum divided into channels
 - Each channel is a narrowband CDMA system



Random Access (RA) MAC Protocols

- Users attempt to access the channel in an uncoordinated manner → Collisions occur
- Collisions detected at destination
 - Destination sends ACKs (acknowledgements)
 - Perfect feedback via ACKs but traffic delay may be high
- Vulnerable period = V_p = time interval during which packets are susceptible to collisions with transmissions from other users

Different RA Approaches

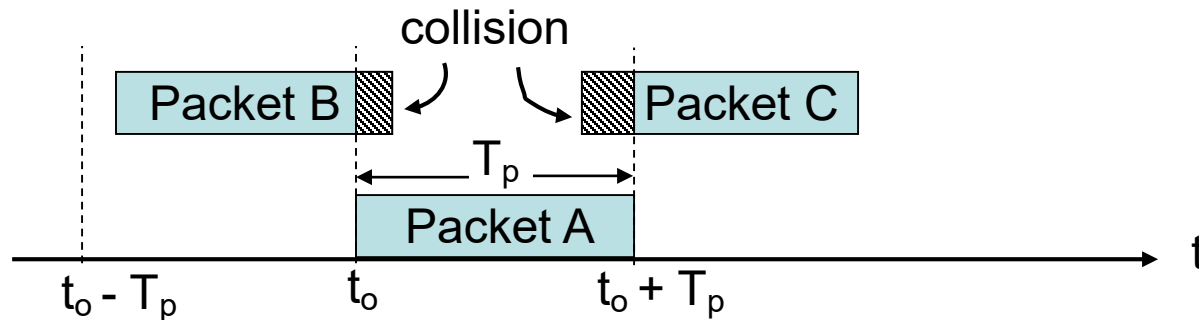
- ALOHA
 - Pure ALOHA
 - Slotted-ALOHA
- CSMA: carrier sense multiple access
 - 1-persistent CSMA
 - non-persistent CSMA
 - p-persistent CSMA
- CSMA/CA: CSMA with collision avoidance
 - BTMA: Busy-tone MA
 - DSMA: Data-sense MA

Pure ALOHA

- Data packetized
- Nodes transmit whenever have information to send
- Nodes transmit packets at arbitrary times
- Collisions occur if packet transmissions overlap by any amount of time
 - If Collision → both packets must be retransmitted
- Node waits for an ACK from receiver
 - If no ACK received, packet assumed lost in collision and retransmitted after a random delay (why ?)

Pure ALOHA

- Assume all packets have the same length (L) and require T_p seconds for transmission
- Each packet vulnerable to collisions for time $V_p = ??$
 - Suppose packet A sent at time t_0
 - If packet B sent any time between $t_0 - T_p$ and t_0
→ end of packet B collides with beginning of packet A
 - If packet C sent any time between t_0 and $t_0 + T_p$, start of packet C will collide with end of packet A
 - Total vulnerable interval for packet A is $2T_p$

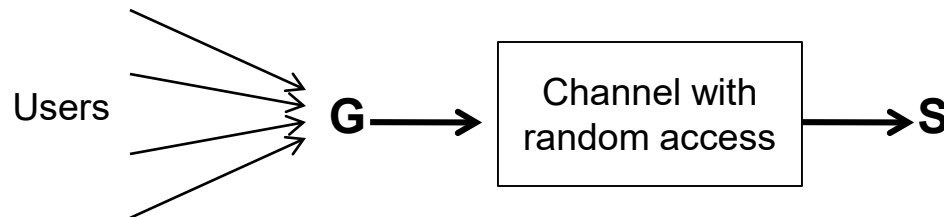


Performance of Random Access

- Performance of fixed-assignment access is measured by:
 - probability of blockage
 - probability of delay if blocked calls are buffered.
- Performance of random access is measured by:
 - channel throughput S and average delay D .

Definitions

- Channel throughput S = average number of successful packet transmissions per time interval T_p
- Offered Traffic G = total traffic offered to the channel = number of packet transmissions attempted per packet time T_p , including new packets as well as retransmissions of old packets
- Delay (D): Average waiting time before successful transmission, normalized to the packet duration T_p

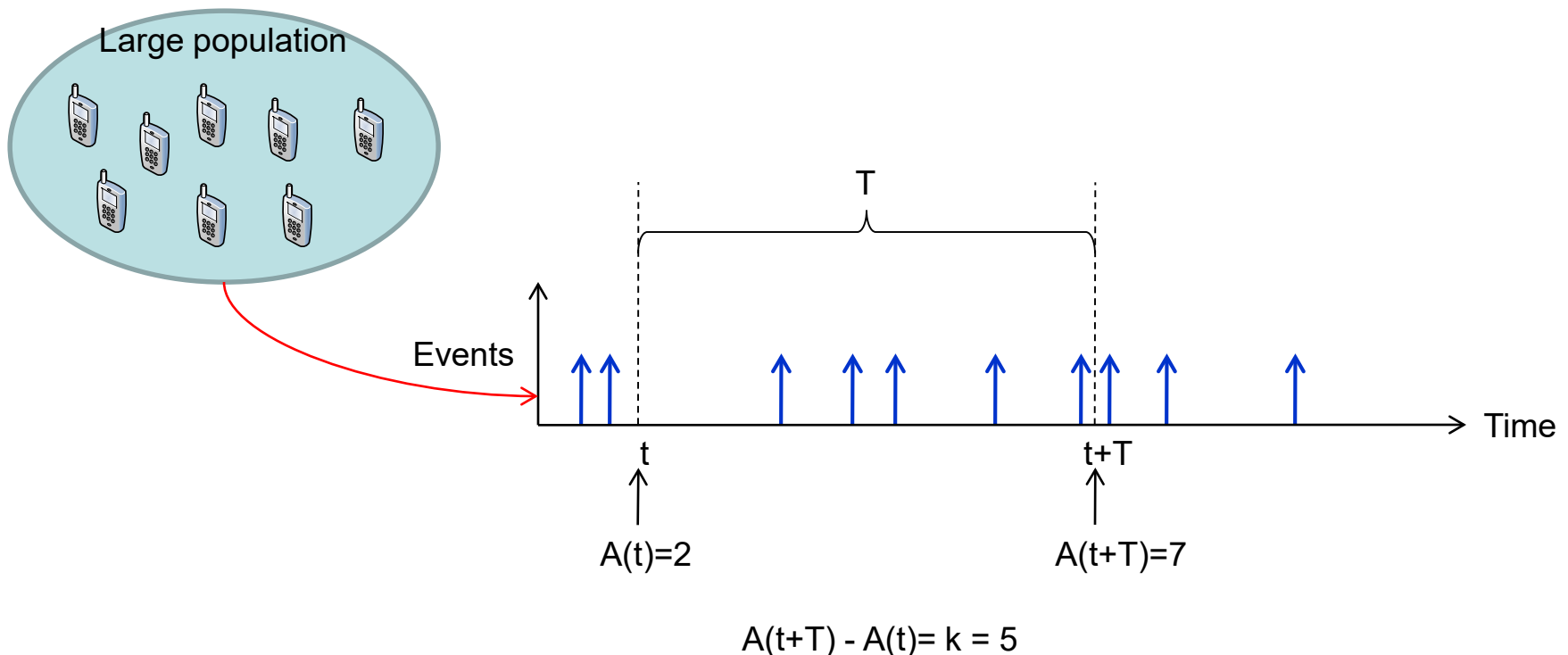


Definitions

- Standard unit of traffic flow is the **Erlang**
 - If channel segmented into intervals of T_p seconds, a traffic flow of one packet per T_p seconds has a value of 1 Erlang
 - Throughput cannot exceed 1 Erlang without collisions
 - $0 \leq S \leq 1$
- If G small, few collisions, few retransmissions, so $S \sim G$
- If G large, many collisions, many retransmissions, so $S \ll G$ and $S \rightarrow 0$

Traffic Model

- Poisson process – a good model for approximating the aggregated traffic of a large number of similar and independent users.



Traffic Model

- Poisson process $\{A(t), t > 0\}$ with mean arrival rate m
 - $A(t)$ is a counting process that represents the total number of arrivals that have occurred from time 0 to t .
 - The probability that k arrivals take place in interval $(t + T) - t$ is given by

$$P\{A(t + T) - A(t) = k\} = e^{-mT} \frac{(mT)^k}{k!}$$

Traffic Model

- Probability that k packets generated (arrive) during given packet time T_p
 - Obeys Poisson distribution with a
 - Mean of m packets per second
 - Mean of G packets per packet period T_p
 - Therefore, $G = mT_p$
- $P(k) = \Pr(k \text{ packets arrive in time } T_p)$

→
$$P(k) = e^{-mT_p} \cdot \frac{(mT_p)^k}{k!} = e^{-G} \cdot \frac{G^k}{k!} = \frac{G^k e^{-G}}{k!}$$

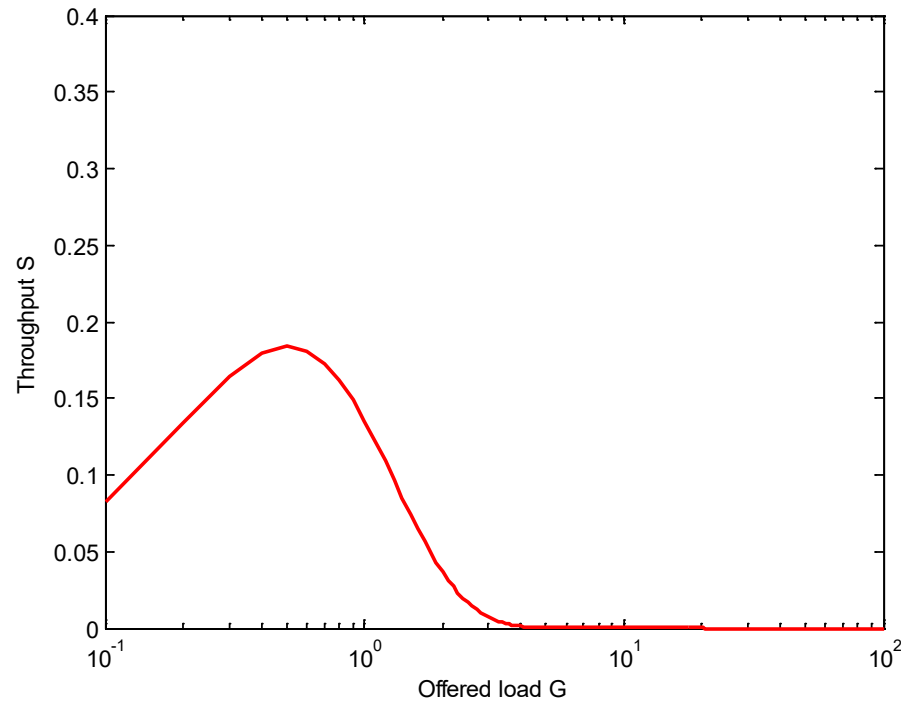
Throughput Analysis of Pure ALOHA

- Throughput $S = G * \text{Pr}(\text{no collisions})$
- For each packet, vulnerable period $V_p = 2T_p$
 - Let's define $P'(k) = \text{Pr}(k \text{ pkts arrive in time } 2T_p)$
 $= \frac{(2G)^k e^{-(2G)}}{k!}$ (why ?)
- Then, $\text{Pr}(\text{no collisions})$
 $= \text{Pr}(\text{no other packet generated during the vulnerable interval} = 2 \text{ packet slots})$
 $= P'(0) = \frac{(2G)^0 e^{-(2G)}}{0!} = e^{-2G}$


- ◆ Therefore, $S = G.P'(0) = Ge^{-2G}$

Throughput of ALOHA

- Throughput, $S = G.P'(0) = Ge^{-2G}$



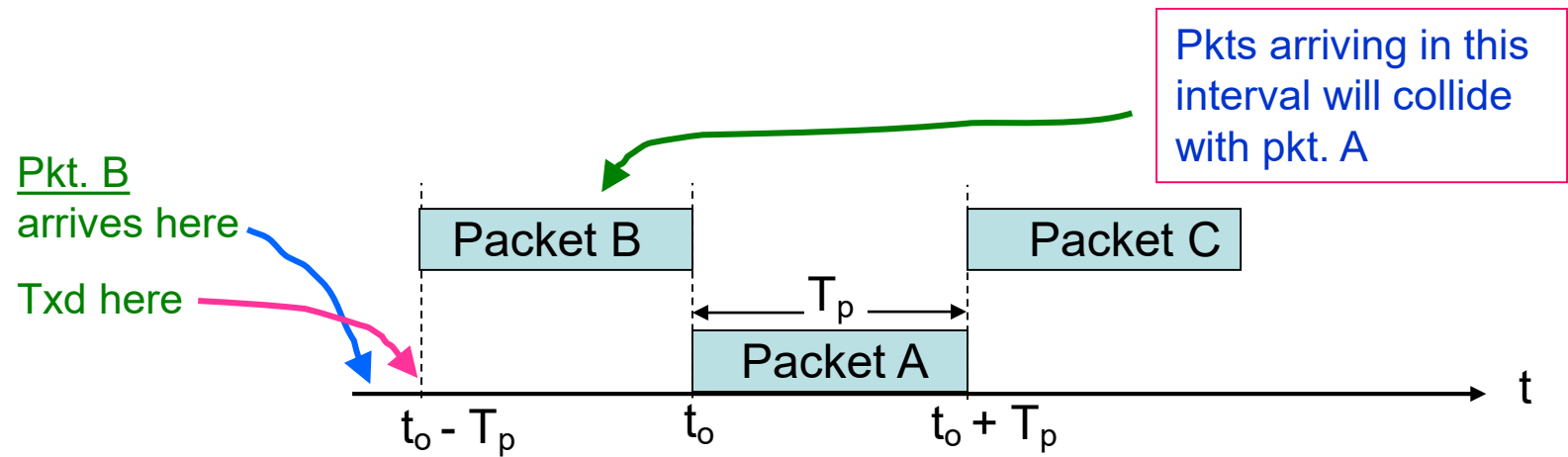
Maximum Throughput

- Max Throughput : at maximum value of S, derivative is 0
 - $dS/dG = e^{-2G} - 2Ge^{-2G} = 0 \rightarrow 1 - 2G = 0$
 - $G_{\max} = 1/2 \rightarrow S_{\max} = Ge^{-2G} = (1/2).e^{-1} = 1/(2e) \sim 0.184$
 - ALOHA can achieve maximum throughput of 18.4%
- Example:
 - System supports 10 Mbps
 - Using pure ALOHA, nodes can get at most 1.84 Mbps of info through network
 - At that peak, total traffic from terminals is
 - $0.5 \times 10 \text{ Mbps} = 5 \text{ Mbps}$
- 1.84 Mbps is successfully delivered pkts (old and new)
- 3.16 Mbps is packets that suffer collisions

Slotted ALOHA

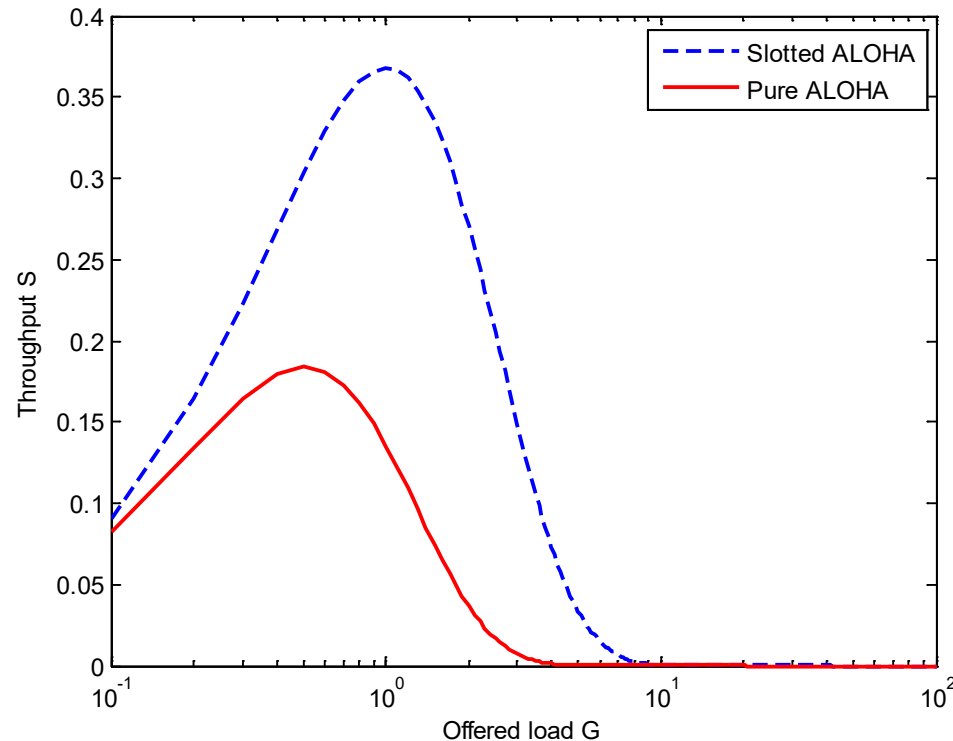
- Can increase efficiency of ALOHA using slotted system
- Transmission time divided into time slots, each slot equal to packet Tx time
- All users synchronized to these time slots
 - Packets held until next time slot for transmission if generated in-between transmission slots
 - Synchronization achieved by transmitting periodic synch pulses from one designated station in the network
- Vulnerable period reduced to T_p

Slotted ALOHA



Throughput of Slotted ALOHA

- $\Pr(\text{no collisions}) = \Pr(\text{no other pkts generated during } T_p)$
 $= P(0) = e^{-G}$
- $S = Ge^{-G}$



Maximum Throughput

- $dS/dG = e^{-G} - Ge^{-G} = 0$
- $G_{\max} = 1.0 \rightarrow S_{\max} = 1/e \sim 0.368$
 - 2x $S_{\max}$ for pure ALOHA
 - 37% of slots have successful data transmissions
- $G = 1 \rightarrow \Pr(0 \text{ packets offered}) = e^{-G} = 1/e \sim 0.368$
 - 37% of slots are empty
 - 26% of slots are in collision
- At higher traffic loads, number of successful slots and empty slots decrease and number of collision slots increases
- Slotted ALOHA is also not efficient

Carrier Sense Multiple Access (CSMA)

- ALOHA is an unstable protocol
 - If G increases to greater than 1 due to fluctuation in offered load, S will decrease
 - Reduction in throughput means fewer successful packet transmissions and more collisions
 - Number of retransmissions increases, backlogging messages to be transmitted and traffic load G increases further
 - This in turn decreases S
 - Results in operating point moving to right and $S \rightarrow 0$
- Random access protocols can be made stable using backoff parameters

CSMA

- ALOHA is inefficient because
 - Users do not take note of transmissions of other users
 - Leads to high collision probability
 - Collisions can be reduced by reducing offered load
 - Leaves channel underutilized
- More efficient approach: users listen to the channel before attempting to transmit a packet
 - This is the basis for CSMA or “listen-before-talk” protocols

CSMA

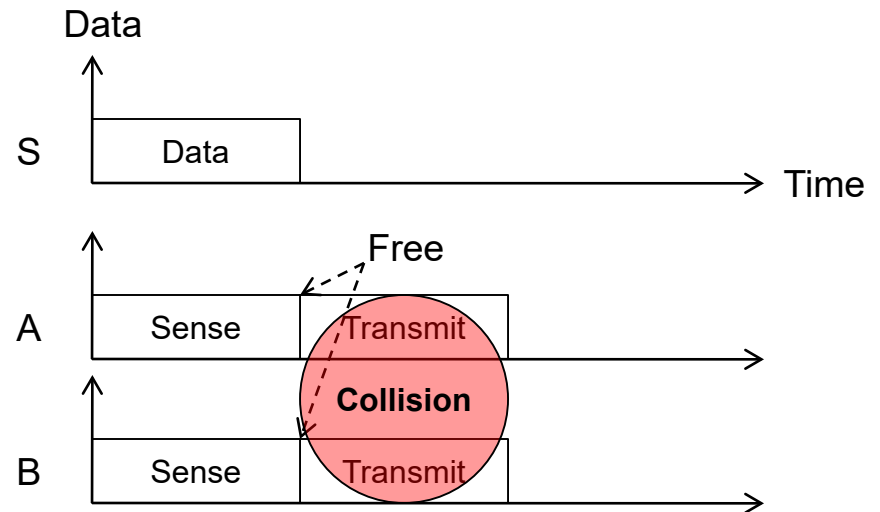
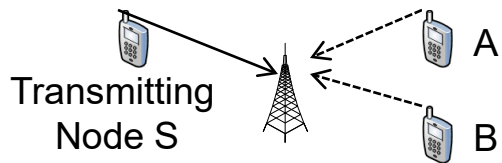
- Detection and propagation delay important parameters
 - Detection delay = time required for node to sense whether or not channel is idle
 - Propagation delay = time required for a single bit to travel from one node to the furthest node on the network
 - Want both small detection time and small propagation delay to minimize collisions
- Different versions of CSMA
 - 1-persistent CSMA
 - non-persistent CSMA
 - p-persistent CSMA

1-persistent CSMA

- Node listens to channel to determine if idle or busy
- If channel is busy, node listens continuously, waiting until the channel becomes free
- Once channel is free, node sends packet immediately
- **Strategy:** transmit packet with $P_r = 1$ when channel is free
- Node waits for an ACK
- If no ACK received, node waits a random amount of time and resumes listening to the channel
- When channel again sensed idle, packet immediately retransmitted

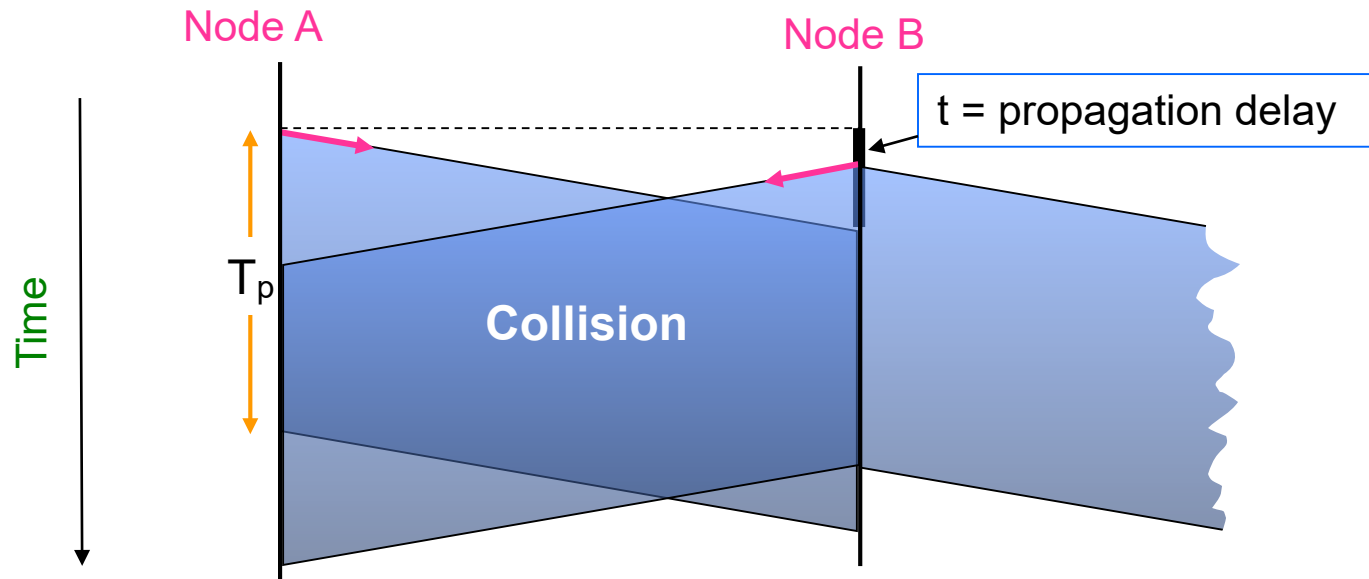
1-persistent CSMA (cont.)

- Collisions can occur because of following causes
 - If Node A and Node B are sensing a busy channel at the same time, as soon as the channel is free, both A and B will begin transmitting their data



1-persistent CSMA (cont.)

- Collisions can occur because of following causes
 - Propagation delays: Node B might sense idle channel even though Node A already began transmitting

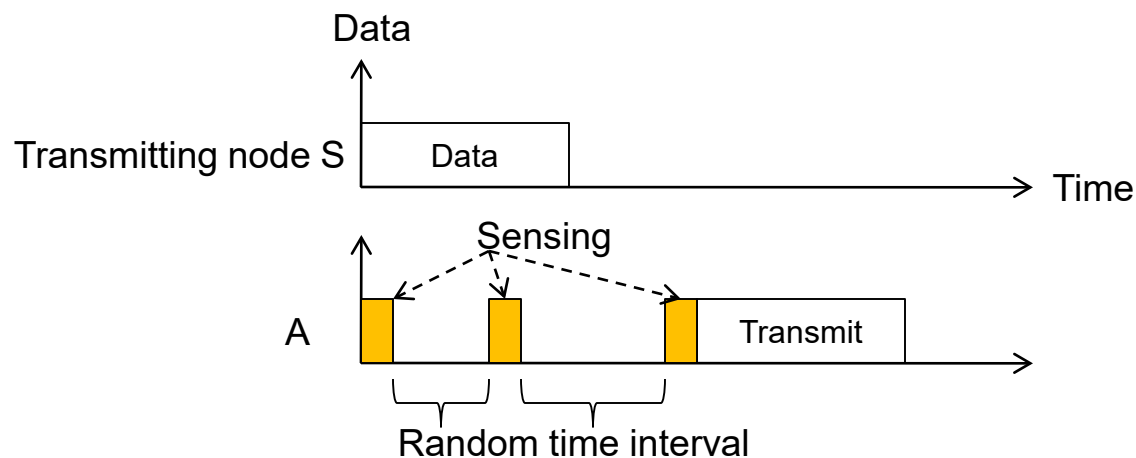


1-persistent CSMA (cont.)

- Throughput performance much better than ALOHA
- Slotted 1-persistent CSMA → where nodes begin transmission at slot boundaries only

Non-persistent CSMA

- Node listens to channel
- If channel idle, node transmits data
- **If channel is busy, node waits a randomly selected interval of time before sensing again**
 - If channel is idle, node transmits data immediately
 - Otherwise, node again waits a randomly selected interval of time before sensing again



Non-persistent CSMA (cont.)

- Randomized waiting times between channel sensings eliminate most collisions resulting from multiple users transmitting simultaneously upon sensing the transition from busy to idle
- Throughput values much higher than 1-persistent CSMA at high traffic loads
 - Max throughput values of 80% or higher
 - At low traffic loads, throughput poor because waiting strategy does not give benefit when there are few users trying to transmit
- Slotted version of non-persistent CSMA also possible

p-persistent CSMA

- Slotted channels where slot length typically chosen to be maximum propagation delay
- Node senses the channel
 - If the channel is idle, node transmits with probability p
 - With probability $1-p$ the node defers action to the next slot, where it senses the channel again
 - if that slot is idle, the station transmits with probability p or defers again with probability $1-p$
 - Procedure repeated until either the packet transmitted or the channel sensed to be busy
 - When channel detected busy, node senses continuously
 - When the channel again becomes free, the node starts the procedure again

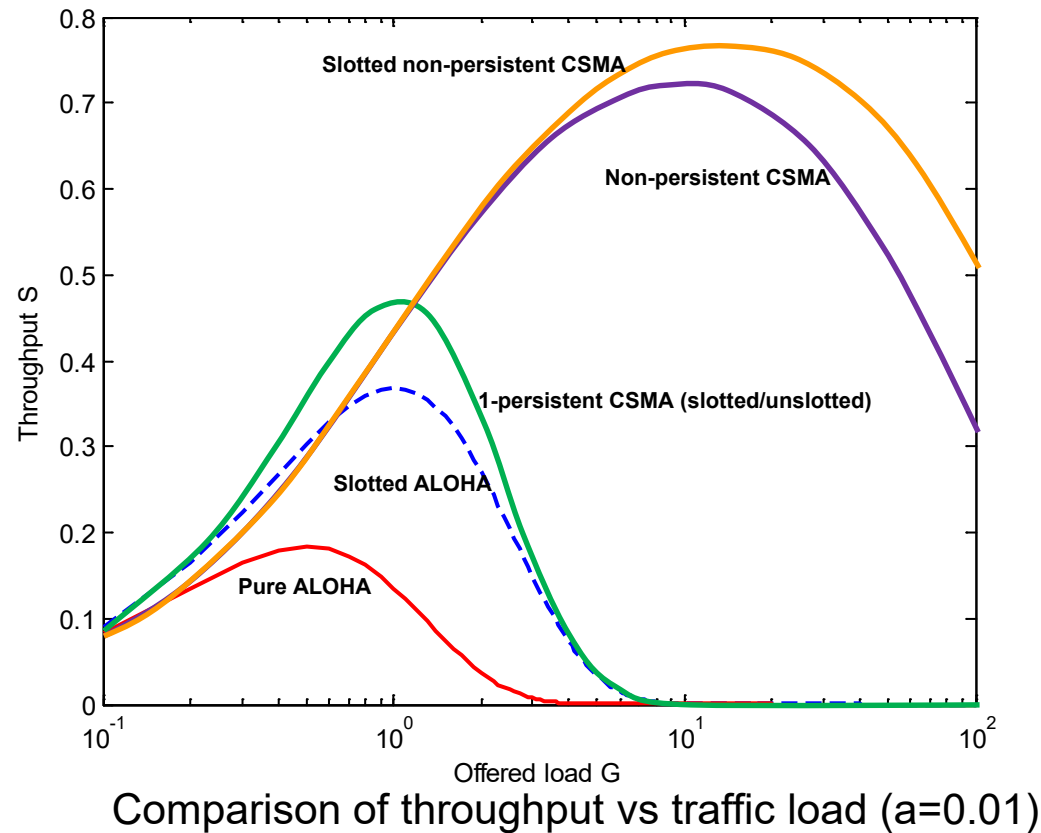
p-persistent CSMA

- For low/intermediate values of propagation delay and with p optimized, throughput of p-persistent CSMA lies between that of slotted and unslotted non-persistent CSMA
- For long propagation delays, p-persistent CSMA throughput is higher than that of non-persistent CSMA

Comparison

- Throughput performance
 - For CSMA protocols, S is a function of $a = t / T_p$ where
 - T_p = pkt. transmission time and
 - t = propagation delay
 - a is the time interval, normalized to packet duration time, during which a transmitted packet can suffer a collision
 - Usually $t \ll T_p$ (since, signal travels at speed of light)
- e.g., $t = d \text{ m} / 3 \times 10^8 \text{ m/s} = 3.33 \text{ microsec/km}$ so $a \ll 1$

Comparison



Comparison

- Slotted and unslotted 1-persistent CSMA essentially indistinguishable
- For low levels of traffic, persistent protocols provide best throughput
- For high levels of traffic load, non-persistent protocols are by far the best
- Slotted non-persistent CSMA has a peak throughput almost twice that of persistent CSMA
- Capacity = peak value of S over entire range of offered traffic load G

Comparison

- For CSMA protocols, capacity is a function of 'a'
 - For large 'a', ALOHA has better capacity
 - With long propagation delay relative to packet transmission time, channel state information arrives too late to be used effectively in reducing collisions → large time in which sender's packet is vulnerable to collisions
- Typically want $a < 0.01$ for good performance using CSMA protocols
 - If max distance between nodes is 100 m, $t = 0.33$ microsec
 - T_p should be greater than 33 microsec to keep $a < 0.01$
 - If datarate, $R_b = 1$ Mbps → minimum packet length = 33 bits
 - If $R_b = 10$ Mbps → minimum packet length = 333 bits

CSMA/CD: CSMA with Collision Detection (CD)

- Node monitors its own transmission to detect collision
- Node must receive while transmitting—if received data different from transmitted data → collision
- Requires transmitter and receiver capable of “listen-while-talk” operation
 - Easy to perform CD over Ethernet → check voltage levels and if different than what was sent, a collision has occurred
 - Hard to detect collisions in wireless media
 - Transmitter drowns out interfering signal at transmitting node
 - Hidden terminal problem
- Too complex to be implemented in most systems

So, Collision Detection is not used in wireless networks

Problems related to Carrier Sensing

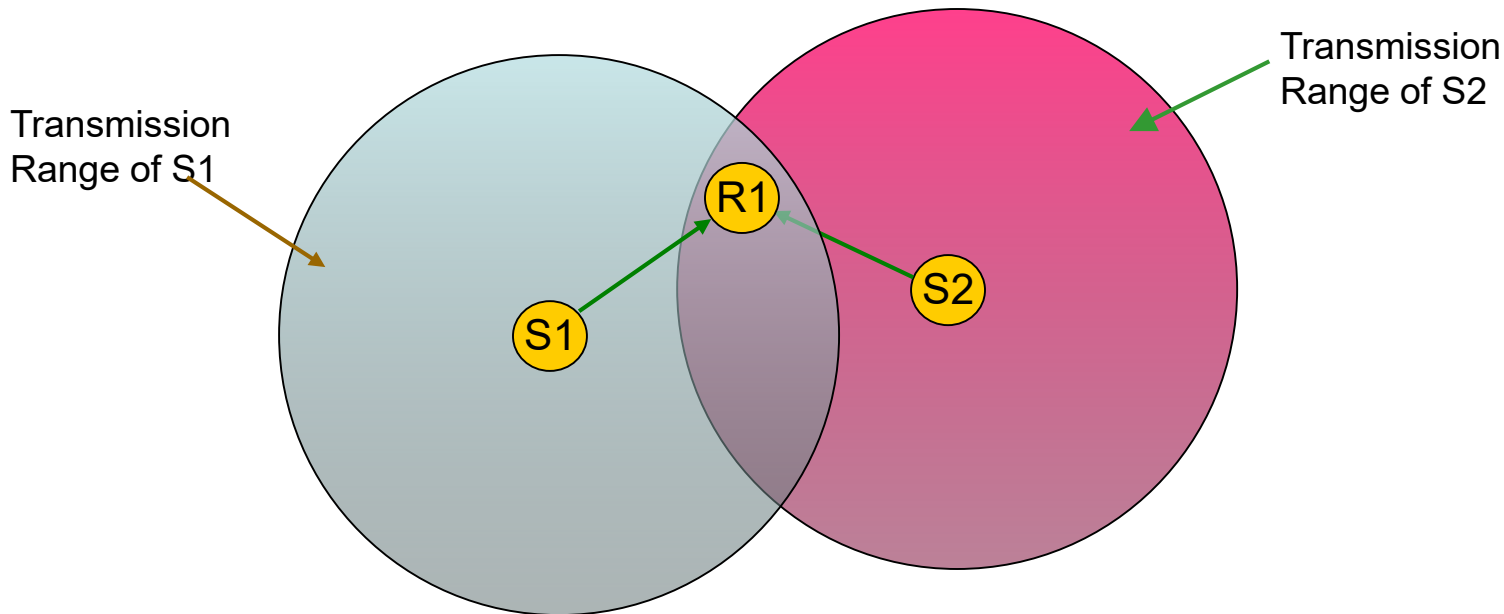
- Carrier Sensing (CS) depends on location of the node
 - Transmitter performs carrier sensing
 - Knows the state of the channel at transmitter
 - Cannot determine the state of the channel at the receiver
- This partial knowledge causes three problems
 - Hidden Nodes
 - Exposed Nodes
 - Capture

Hidden Terminal Problem

- A Hidden Terminal is a node that is within range of the receiver but not within range of the transmitter
- A hidden node will sense an idle channel and transmit data
- This transmission can interfere with the reception at the receiver

Hidden Terminal Problem (contd.)

- Suppose node S1 is transmitting to node R1
- Node S2 will sense an idle channel as it is outside the range of node S1
 - Now, S2 transmits data to node R1
- Causes collision at R1



Hidden Terminal Problem (contd.)

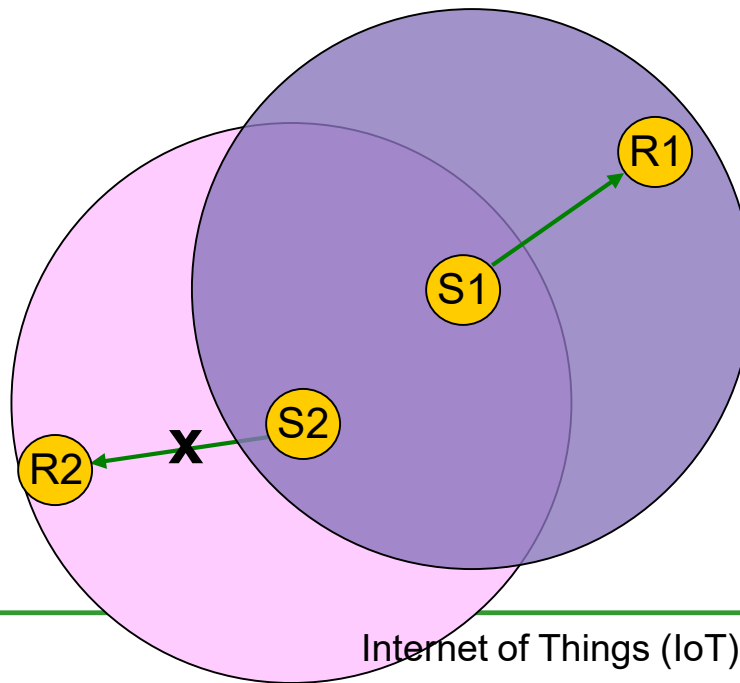
- Problem: collisions must be detected at the transmitter but actually occur at the receiver
- Nodes outside the range of the transmitter “hidden” to the transmitter
→ CSMA not effective

Exposed Terminal Problem

- **Exposed Terminal:** A node that is within range of the sender but out of range of the destination
- An exposed node will sense a busy channel and will not transmit
- Causes underutilization of the channel capacity

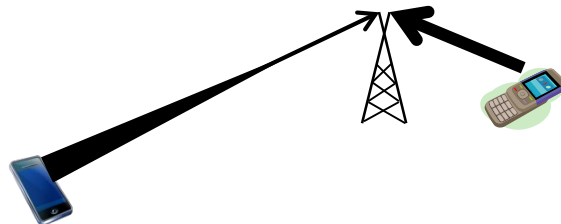
Exposed Terminal Problem (contd.)

- Suppose node S1 is transmitting to node R1
- Node S2 will sense a busy channel
- However, node S2 could transmit to node R2
 - R2 is outside the range of node S1
 - No collision
- Node S2 (exposed node) backs-off (unnecessarily)



Capture Effect

- When receiver can cleanly receive a transmission from one of two simultaneous transmissions
 - Collisions may not cause both packets to be “lost”
 - Strongest user can successfully capture receiver
 - Packet from strongest user may survive collisions
- Often closest user captures receiver due to smaller propagation path loss



Capture Ratio

- Minimum power ratio which one arriving packet must have relative to the other colliding packets in order that it can be received successfully is called **Capture Ratio**
- Capture ratio depends on receiver and modulation method used

Capture Effect

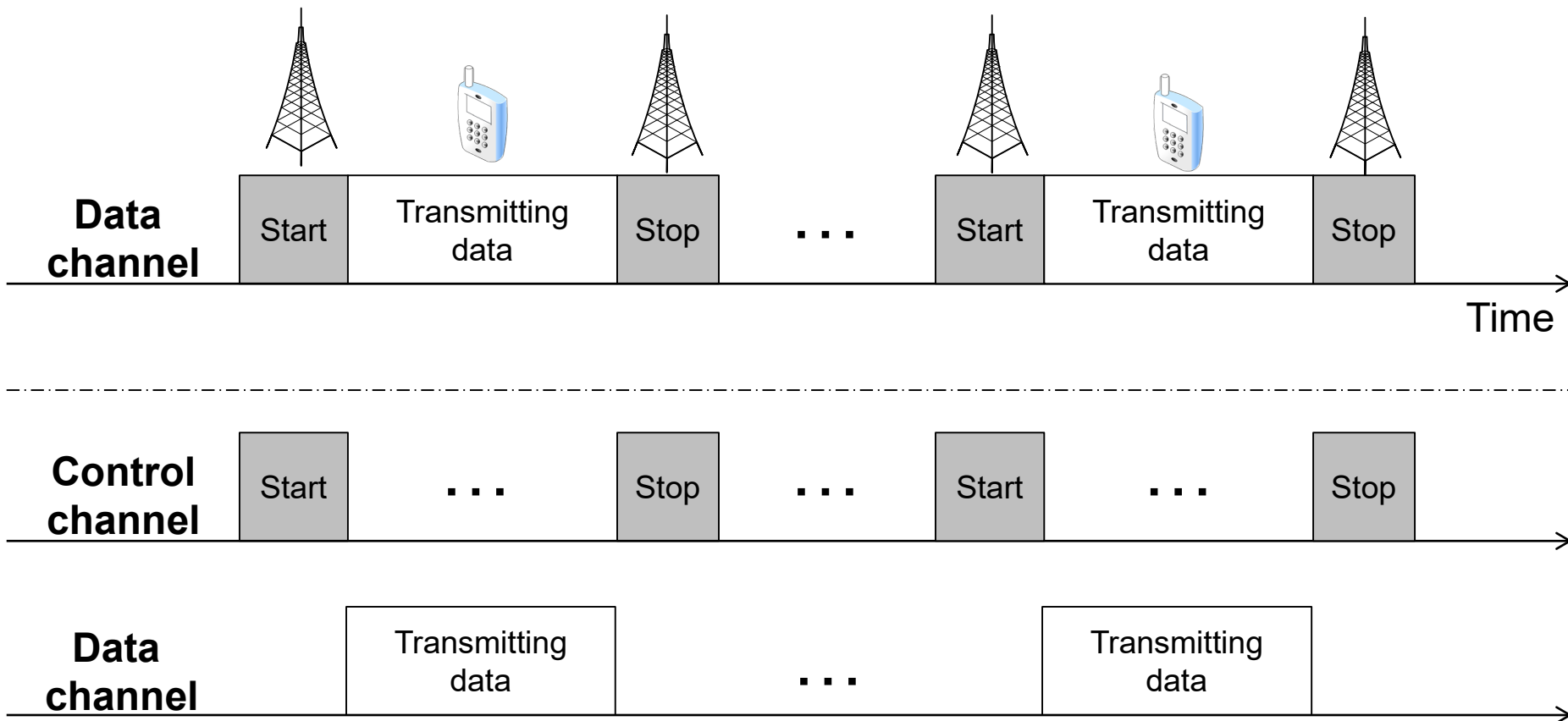
- Capture Effect results in unfair sharing of the bandwidth
 - Near-far effect: Nodes close to the receiver can easily capture the receiver and prevent nodes further away from communicating with the receiver
 - Wireless MACs aim to ensure fairness in the presence of capture

CSMA/CA: CSMA with Collision Avoidance

- Collision avoidance implemented using signaling, backoffs, packet reservation
- CA implemented using
 - Out-of-band signaling
 - Receiving nodes transmit a signal to let nodes in their range know that the channel is busy
 - Eliminates hidden nodes
 - Increases the number of exposed nodes
 - Example: Busy Tone Multiple Access (BTMA)
 - Control handshaking
 - Each stage of the handshake lets nodes know if they are in range of the transmitter, receiver, or both
 - Tradeoff in overhead for handshaking and the number of hidden nodes eliminated
 - Example: MACA, MACAW, etc.

CSMA/CA: CSMA with Collision Avoidance

- Out-of-band communication is the exchange of call control information in a separate band from the data or voice stream, or on an entirely separate, dedicated channel



CSMA with Busy-Tone Signaling (BTMA)

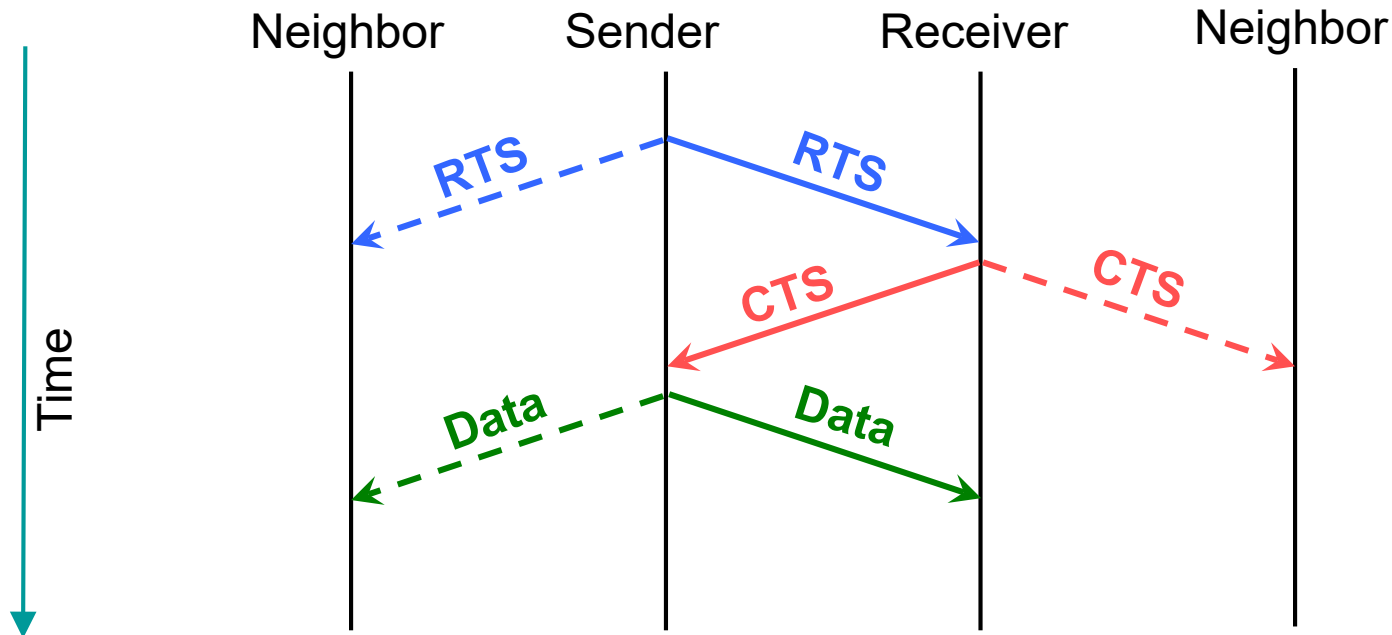
- Whenever a node is receiving message, it can send a signal on the “busy-tone” channel to let all nodes in communication range know that the medium is busy
- In centralized network, BS sends busy-tone signal
- In decentralized network, all nodes must transmit busy-tone signals when they detect busy channel

CSMA with Control Handshaking

(e.g., Multiple Access Collision Avoidance, MACA)

- Problem with CSMA/CA is that transmitter cannot determine state of carrier at receiver
 - Hidden terminal problem
 - Exposed terminal problem
- Protocol implements hand-shaking collision avoidance
 - Precede data transmission with request-to-send (RTS) packet
 - RTS contains length of expected data transmission (all phases)
 - All nodes in vicinity of transmitter node enter backoff for long enough time to allow the sender to receive CTS message, if any
 - If receiver node successfully receives RTS, replies with a clear-to-send (CTS) packet
 - CTS packet contains length of expected data transmission
 - All nodes in vicinity of receiver node enter backoff for duration of message delivery
 - Upon receipt of CTS, transmitter node sends data

CSMA with Control Handshaking



Handshaking Sequence

CSMA with Control Handshaking

- Protocol successfully alleviates the hidden terminal and exposed terminal problems
 - All nodes that could corrupt transmission (any nodes hearing CTS) know not to transmit – This is called: **virtual carrier sensing**
 - All nodes that would not corrupt transmission (any nodes only hearing RTS and not CTS) can transmit without corrupting communication between receiver and transmitter
- Protocol assumes transport layer reliability (e.g., TCP)
 - There is no mechanism at MAC layer to guarantee delivery

CSMA with Control Handshaking

Backoff

- If a node does not receive a CTS in response to an RTS, it will eventually assume that collision occurred
- Other nodes in vicinity also trying to use the channel
- Node backs off before retransmitting RTS to reduce probability of another collision
 - Node chooses a random number x between 1 and BO (Current value of the Back Off window)
 - Node must wait x time slots before retransmitting
- MACA uses **Binary Exponential Backoff (BEB)**
 - If an RTS is unsuccessful (i.e., it does not elicit a CTS), the BO is doubled (up to a maximum value)
 - Otherwise, wait time set to minimum value (of BO window)

CSMA with Control Handshaking

The Significance of RTS/CTS

- Collisions usually occur in RTS/CTS packets
 - If RTS/CTS packets unsuccessful, most nodes know that channel is busy and will defer their own transmissions
 - Some nodes may not receive RTS or CTS due to other transmissions in their vicinity
 - After successful RTS/CTS, the probability that Data will meet with no collision is very high
- RTS/CTS packets short compared to data packets
 - Even if RTS/CTS packets collide, they do not use much of the channel resources
 - Data Packets that do require large amounts of channel resources have a high probability of being successfully transmitted
 - Good policy!

CSMA with Control Handshaking

(e.g. Multiple Access with Collision Avoidance for Wireless, MACAW)

- MACAW has some improvements
 - Additional control message, for example,
 - Acknowledgment (ACK) message
 - Change in the Backoff Counter Update mechanism

Multiple Access with Collision Avoidance for Wireless

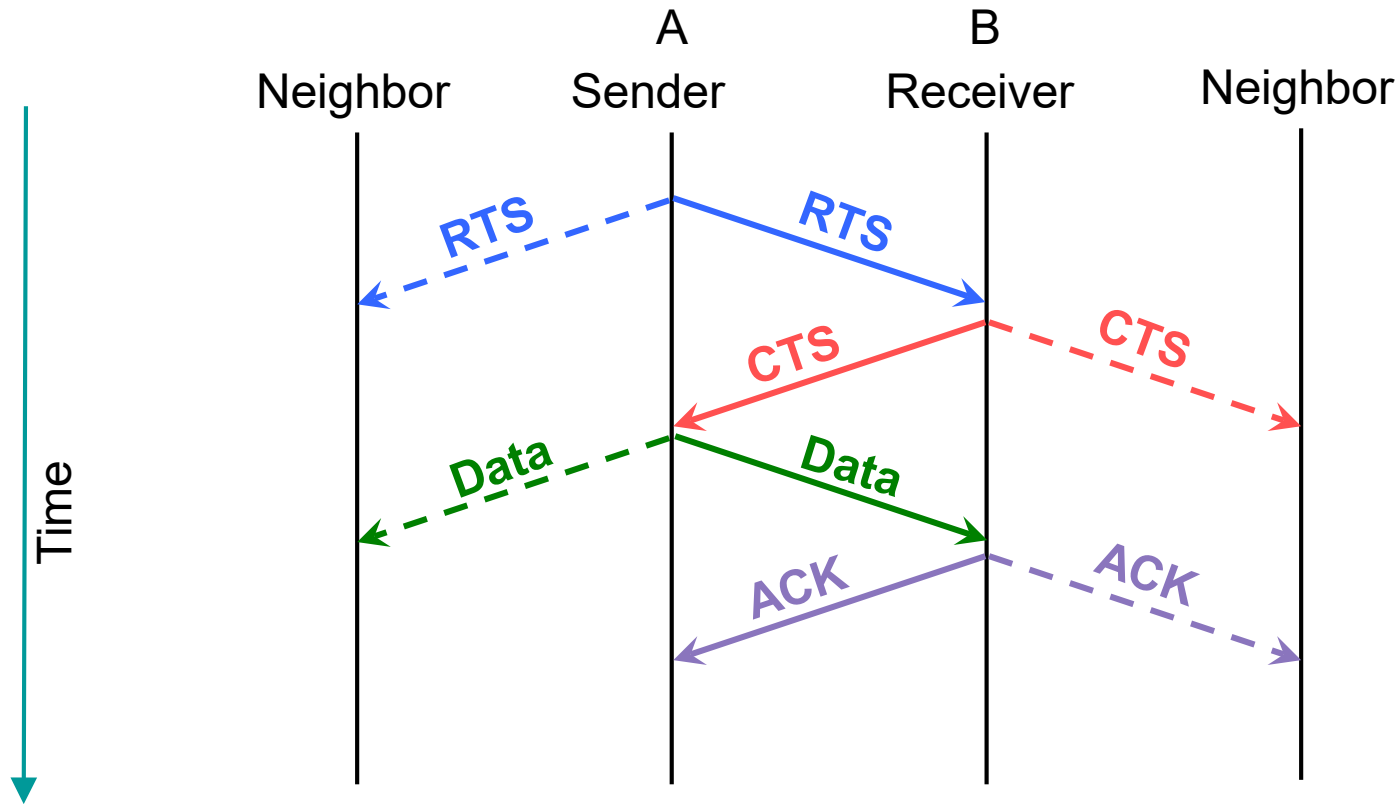
ACK Message

- ACK message is used to add functionality to ensure reliable delivery of data at the link layer

Suppose node A sending data to node B

- Node A sends RTS
- Node B sends CTS
- Node A sends Data
- If data received successfully, node B sends ACK

ACK Message



Handshaking Sequence with ACK

ACK Message

- If node A does not receive ACK after a certain time, node A retransmits RTS
- If node B had received the data
 - ACK message was lost, not data message
 - Node B sends ACK in response to RTS
- If node B did not receive the data
 - Node B sends CTS in response to RTS
 - Data is retransmitted
- Node A's backoff counter increases if initial data transfer was unsuccessful to minimize future collisions

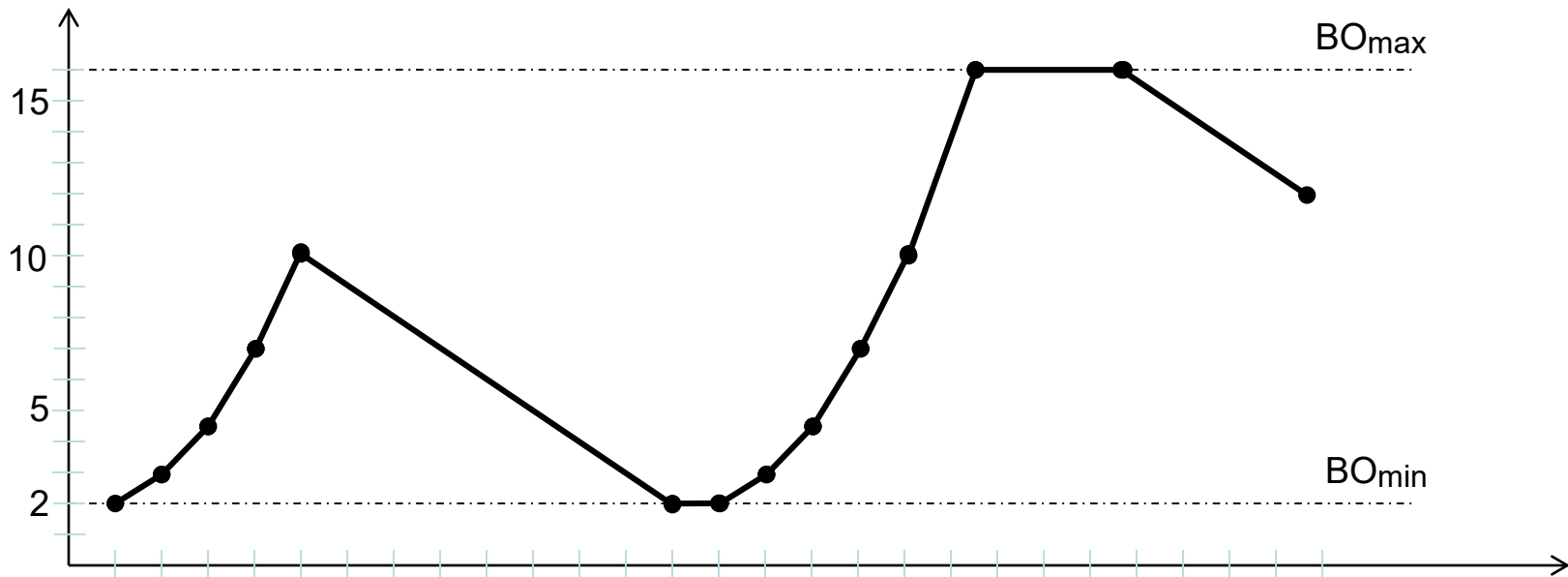
Backoff

- Binary Exponential Backoff (BEB) can result in one stream starving another
 - With every unsuccessful RTS, node will backoff with value “BO”
 - Node with successful RTS will have backoff (BO_{min}) at minimum value
 - Node with unsuccessful RTS will have backoff (BO_{max}) at maximum value
 - If no maximum backoff value, one node would eventually have infinite backoff
 - Other node would permanently capture channel
 - High throughput but not fair allocation
- All packet headers are modified to include node's backoff value
 - Nodes receiving packet copy this BO value
 - All nodes will have equal BO values

Backoff

- Backoff algorithm modified to multiplicative increase, linear decrease
 - Upon collision, BO increased to $\min(1.5 \cdot \text{BO}, \text{BO}_{\max})$
 - Upon success, BO decreased to $\max(\text{BO}-1, \text{BO}_{\min})$

Backoff



Power Consumption of MAC Protocols

- Radio operates in 3 modes:
 1. Transmit (Tx)
 2. Receive (Rx)
 3. Standby (SB)
- Relative powers dissipated in three modes
 - $P_{TX} > P_{RX} \gg P_{SB}$ for long-range communication
 - $P_{TX} \sim P_{RX} > P_{SB}$ for short-range, low power transceivers
 - Different MAC protocols will be “low-power” depending on model of transceiver power dissipation
 - Switching between states causes time delay and power dissipation

How to Minimize Energy Consumption ?

- Place nodes in standby mode as much as possible
 - Nodes do not need to be 'ON' when not receiving data
 - Requires nodes to know when they must listen to the channel and when they can "sleep"
 - Node must know when other nodes have data to transmit to it
- Minimize Collisions
 - Reduce retransmissions as they are costly in terms of energy
 - Introduce delays (e.g. random backoff)
 - Use contention for reservations and reservations (no contention) for data transmission

How to Minimize Energy Consumption ?

- Allocate contiguous slots for transmission/reception
 - Avoids power/time in switching from Tx to Rx
- Have the BS buffer packets for a node and transmit all packets at once
 - Allows node to remain asleep for long time
 - Trade-off in delay to receive packets and BS buffer size
- Tradeoff energy consumption and delay in receiving a message
 - Easier to do in Centralized networks (as above) than in Distributed networks
- Centralized scheduling is most energy-efficient

Summary

- Problems of Carrier Sensing based Random Access
 - Hidden Terminal
 - Exposed Terminal
 - Capture
- Solutions:
 - Out-of-band signalling, e.g.: BTMA
 - Control Handshaking: E.g. MACA, MACAW
- Random Access Protocols
 - MACA
 - MACAW
- Controlled Random Access Protocols
 - Use Contention for reservation of channels
 - D-PRMA
 - Other protocols